C3103

Computer Networks

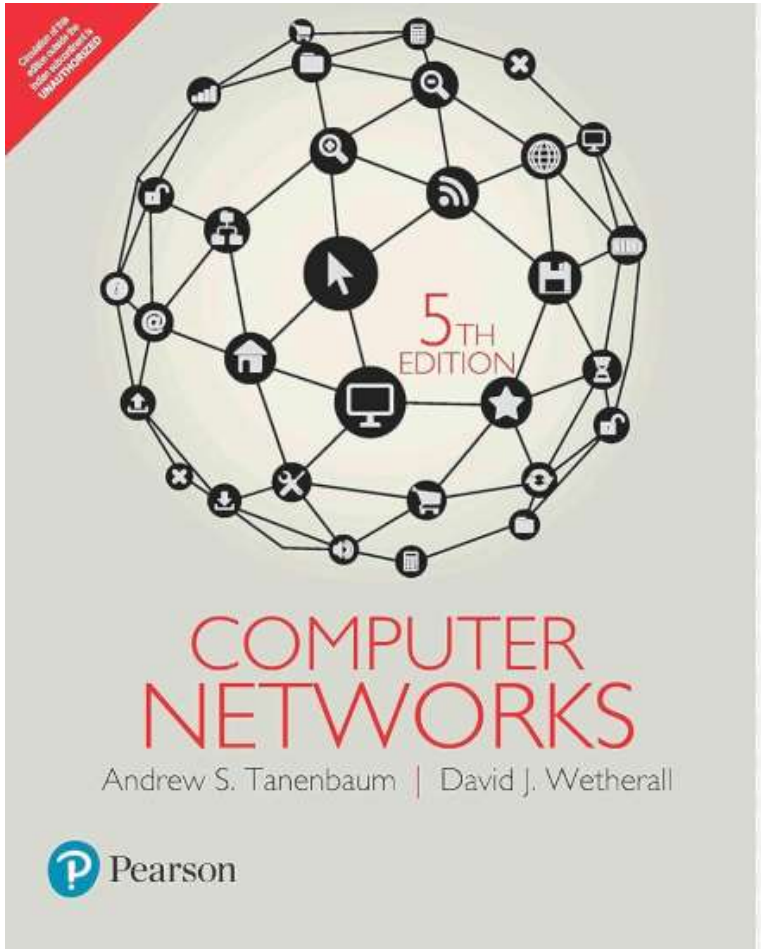


Dr. Pradeepthi K V

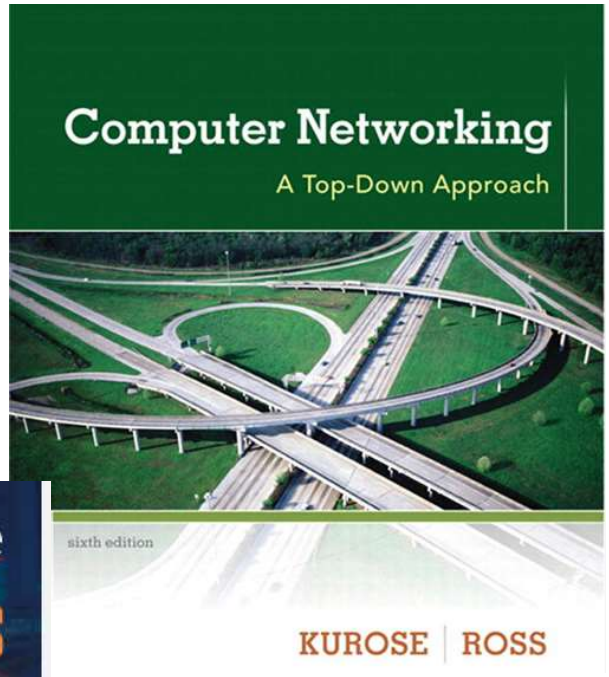
C R Rao Advanced Institute of Mathematics, Statistics and Computer Science

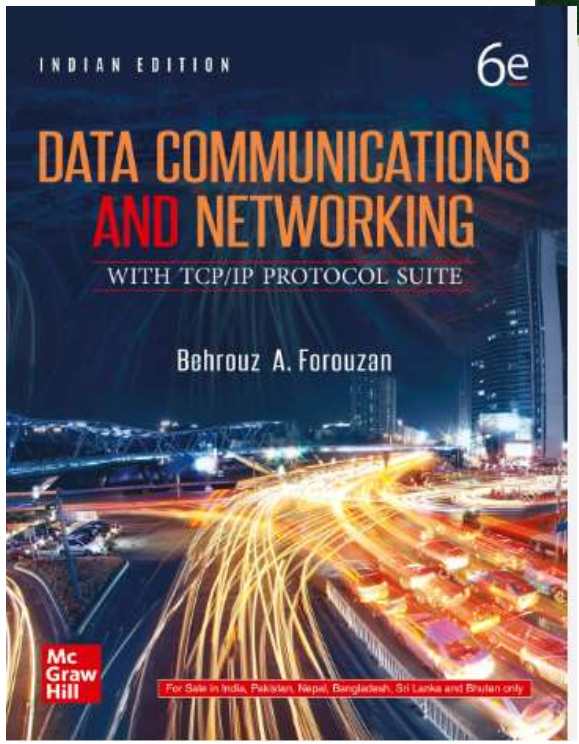
University of Hyderabad Campus, Gachibowli, Hyderabad.

Text book as per JNTU



Reference Book 1





Reference Book 2

Unit-1

- Network hardware, Network software, OSI, TCP/IP Reference models, Example Networks: ARPANET, Internet.
- Physical Layer: Guided Transmission media: twisted pairs, coaxial cable, fiber optics, Wireless Transmission.
- Data link layer: Design issues, framing, Error detection and correction.

- Reference:

Computer Networks, 5th Edition by Andrew Tanenbaum & David Wetherall, Copyright. Pearson Education 2011

Introduction

- What is Internet?
- What are Protocols?
- What are RFCs?

Need for a Standard

- Recognize data
- Divide data into manageable chunks
- Add information to each chunk to:
  - Determine where the data is coming from
  - Identify where the data is going to
- Add timing and error checking information
- Put the data on the network and send it on its way

Standard Protocols Needed

- Network operating systems follow strict rules (protocols) to control how each of the previous tasks are accomplished.

- Need for standard protocols so that software/hardware from different vendors could communicate

Standards Organizations

- ISO—International Organization for Standardization
  - Collection of organizations from 130 countries
  - Establish international technological standards to facilitate global exchange of data

- ITU—International Telecommunication Union
  - United Nations agency that regulates international telecommunications including radio and TV frequencies, satellite and telephone specifications, and networking infrastructure

Social Issues

- Network neutrality – no network restrictions
- Content ownership, e.g., DMCA takedowns
- Anonymity and censorship
- Privacy, e.g., Web tracking and profiling; terrorists
- Jurisdiction: gambling, political speech,etc
- Theft, e.g., botnets and phishing
- Incorrect (medical) information and fake news
- Facebook & Twitter and Russian trolls

Network Hardware

Terminology :

- 2 types of Transmission Technology
  - ❑ Broadcast links
    - ❑Multi cast
  - ❑Point-to-points
    - ❑Uni cast
- Packets
- Bandwidth
- Tranmission Rate

Network Hardware (1)

- Personal area networks
- Local area networks
- Metropolitan area networks
- Wide area networks
- The Internet

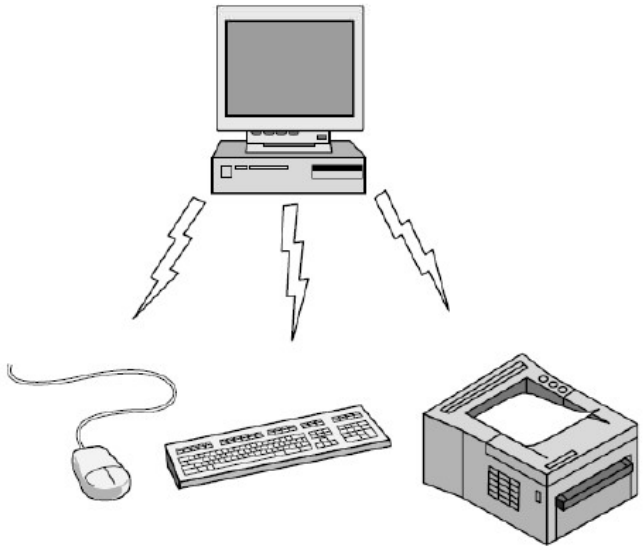
Network Hardware (2)

Classification of interconnected processors by scale.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

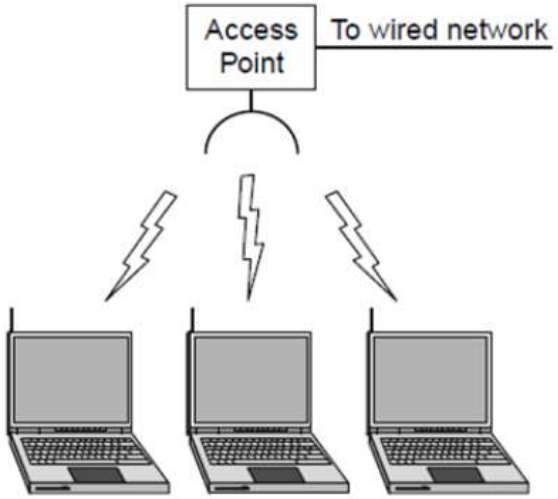
Personal Area Network

- Connect devices over the range of a person
- Example of a Bluetooth (wireless) PAN:

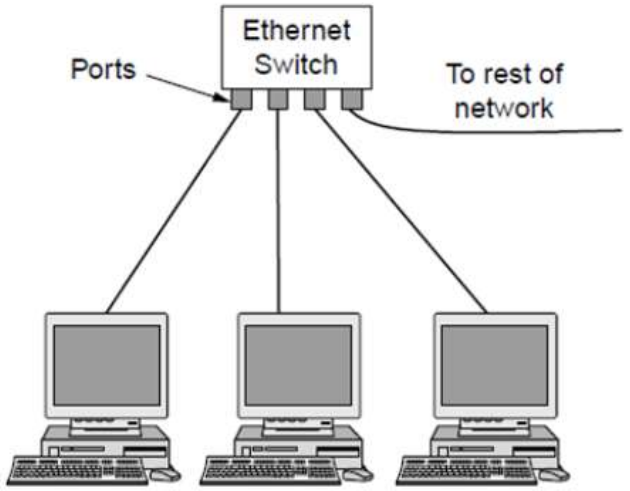


Local Area Networks

- Connect devices in a home or office building
- Called enterprise network in a company



Wireless LAN
with 802.11/a/b/g/n/ac/ad

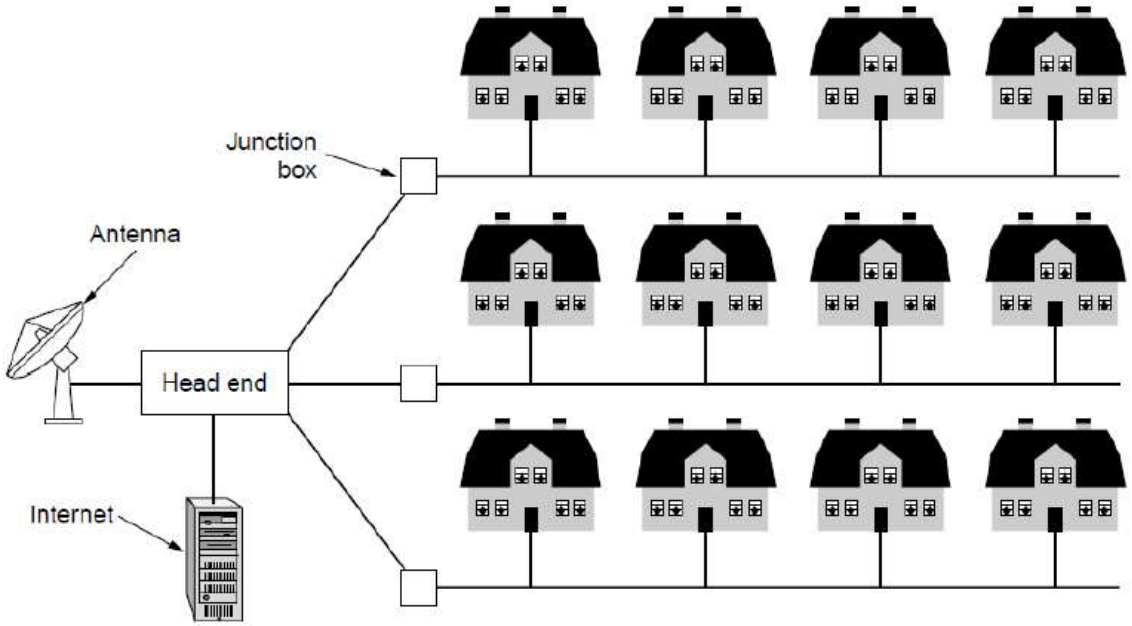


Wired LAN with
switched Ethernet

- Ethernet
- Switch
- Port

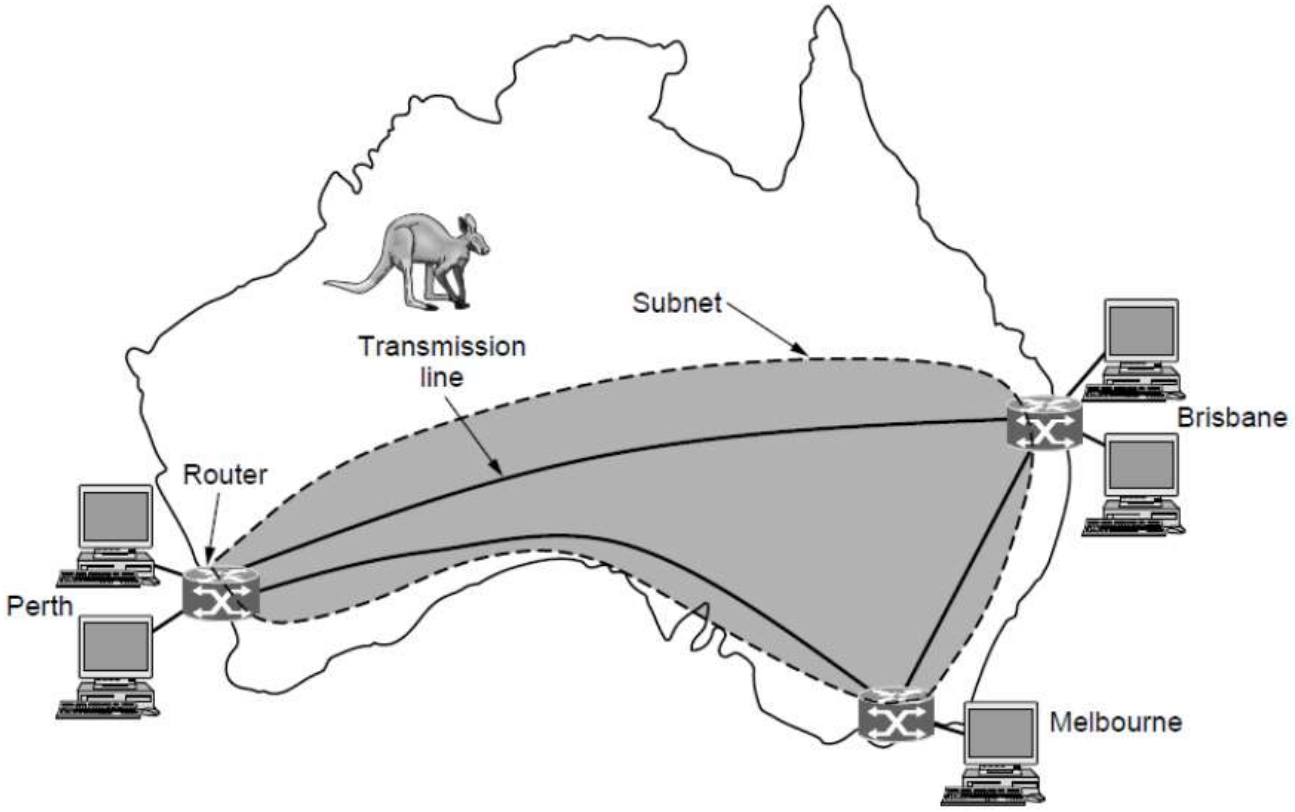
Metropolitan Area Networks

- Connect devices over a metropolitan area
- Example MAN based on cable TV:



Wide Area Networks (1)

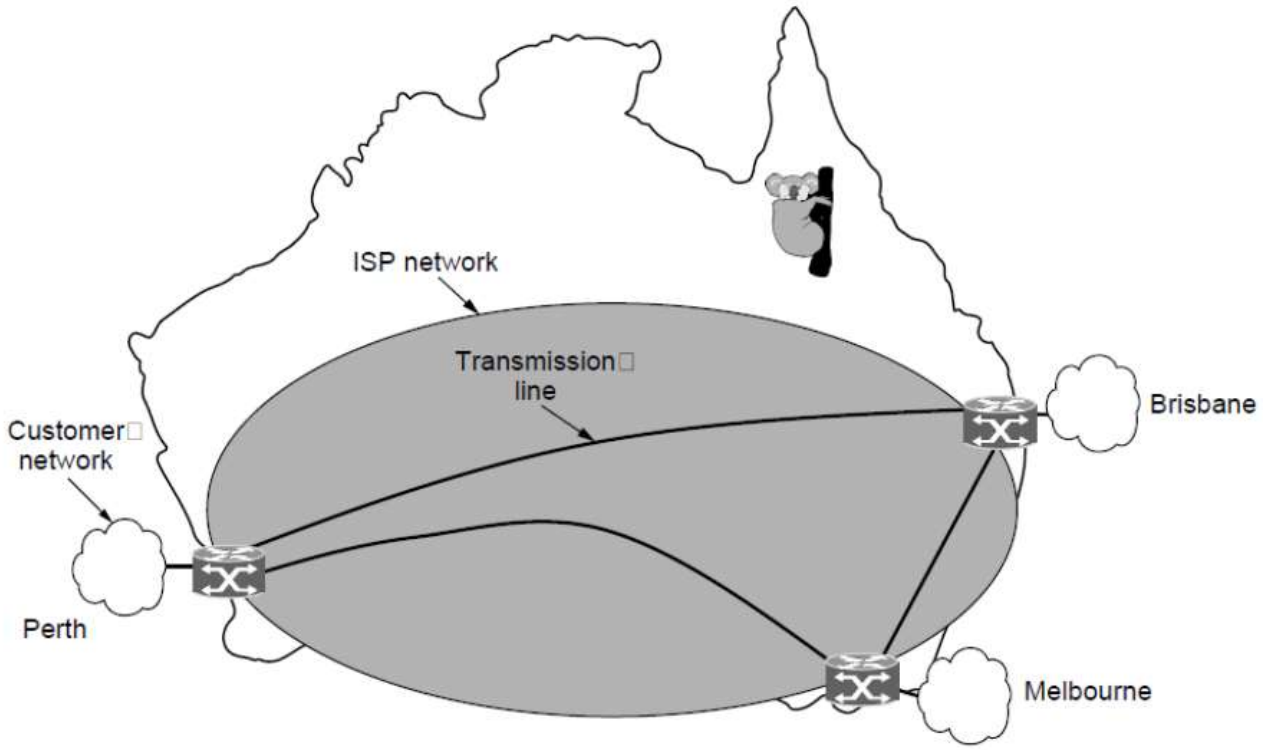
- Connect devices over a country
- Example WAN connecting three branch offices:



- Router
- Switch
- ISP-Internet Service Provider
- Routing Algorithm
- Forwarding Algorithm

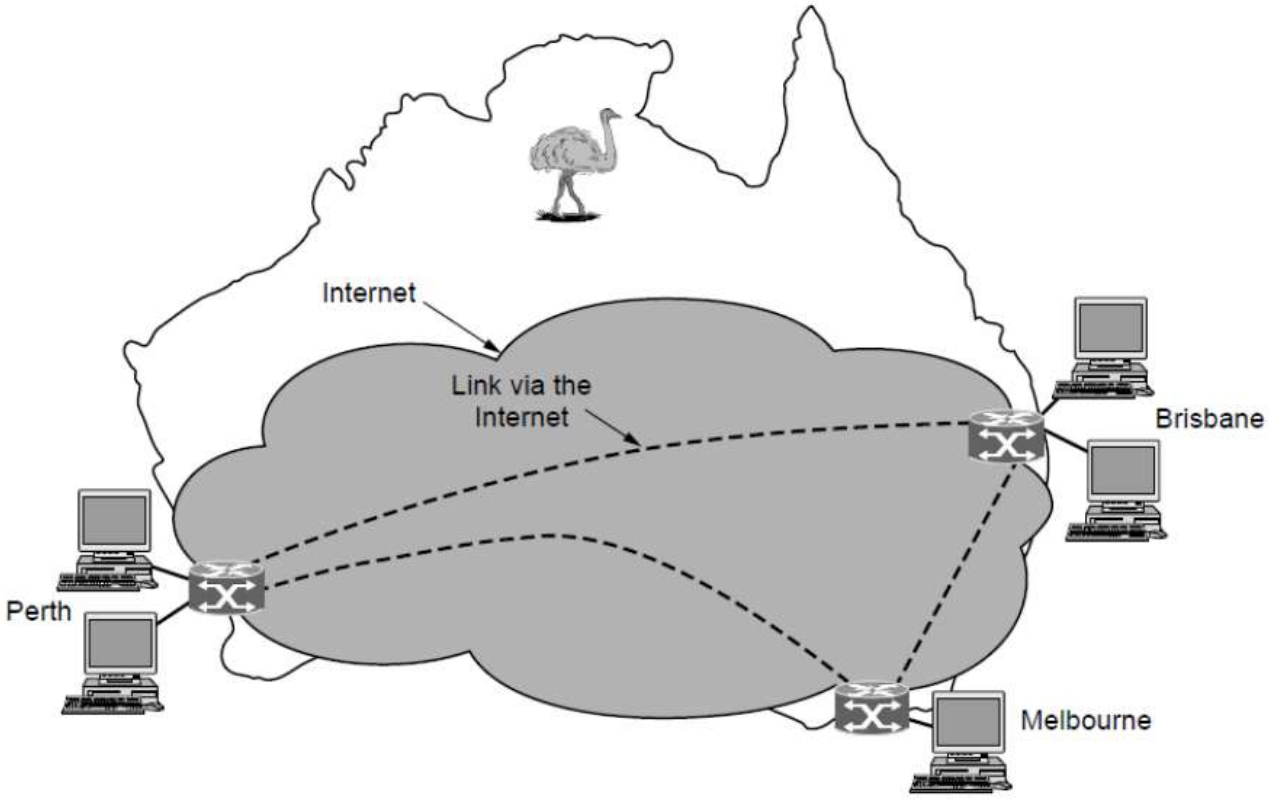
Wide Area Networks (2)

- An ISP (Internet Service Provider) network = WAN.
- Customers buy connectivity from the ISP to use it.



Wide Area Networks (3)

A VPN (Virtual Private Network) is a WAN built from virtual links that run on top of the Internet.



Network Software

- Layers and abstractions
- Peers and protocols
- Design issues for the layers
- Connection-oriented vs. connectionless service
- Service primitives
- Relationship of services to protocols

Common Protocols You May Know

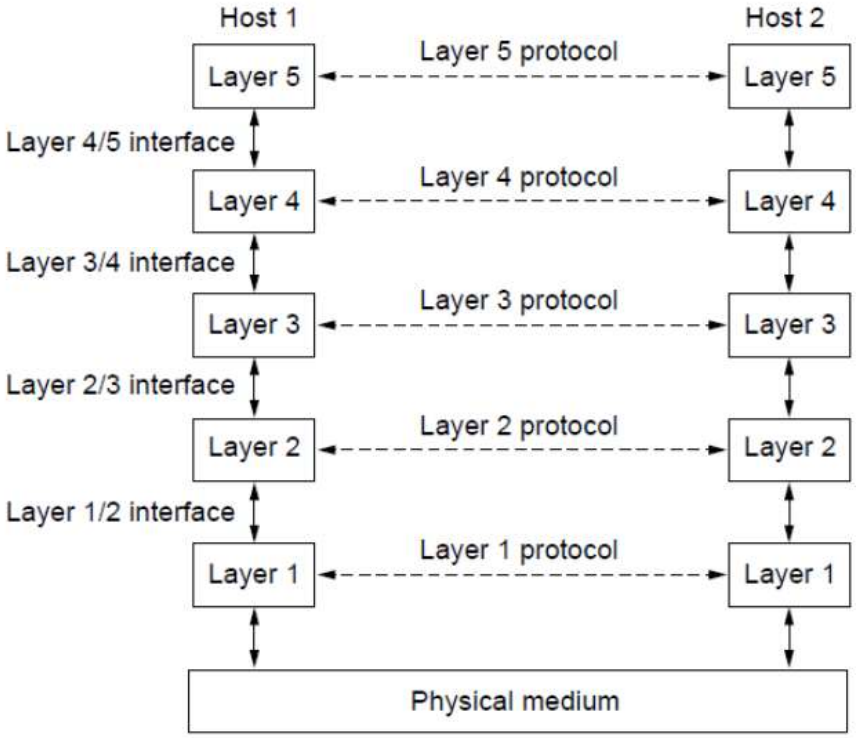
- TCP
- IP
- 802.11
- HTTP
- SSL
- DNS

Protocols define the format, order of messages sent and received among network entities, and actions taken on message transmission, receipt

Protocol Layers (1)

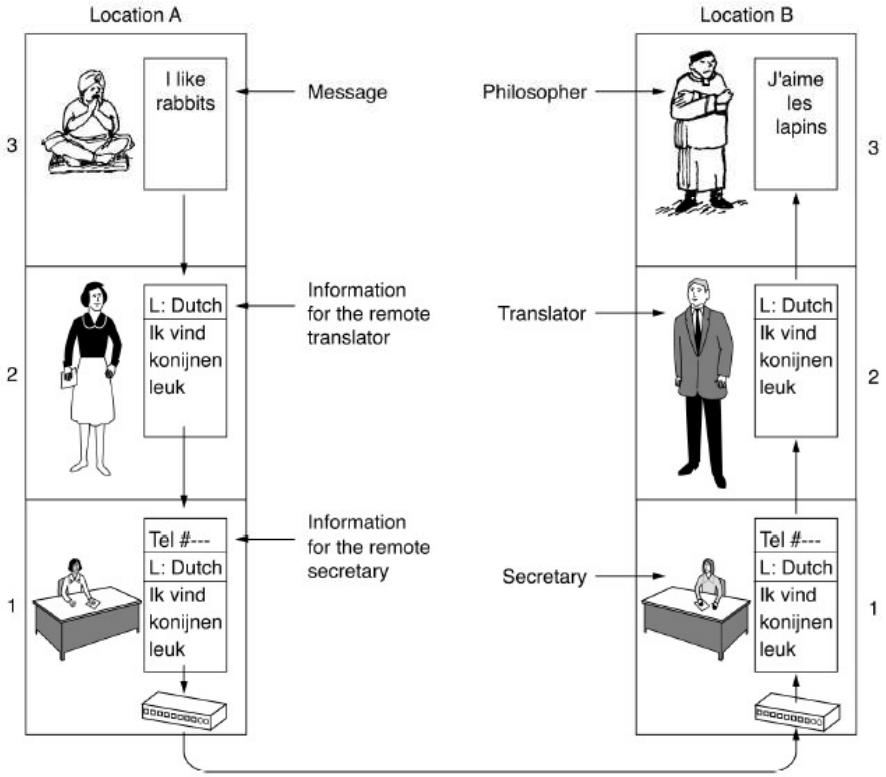
Protocol layering is the main structuring method used to divide up network functionality.

- Each protocol instance talks virtually to its peer
- Each layer communicates only by using the one below
- Lower layer services are accessed by an interface
- At bottom, messages are carried by the medium



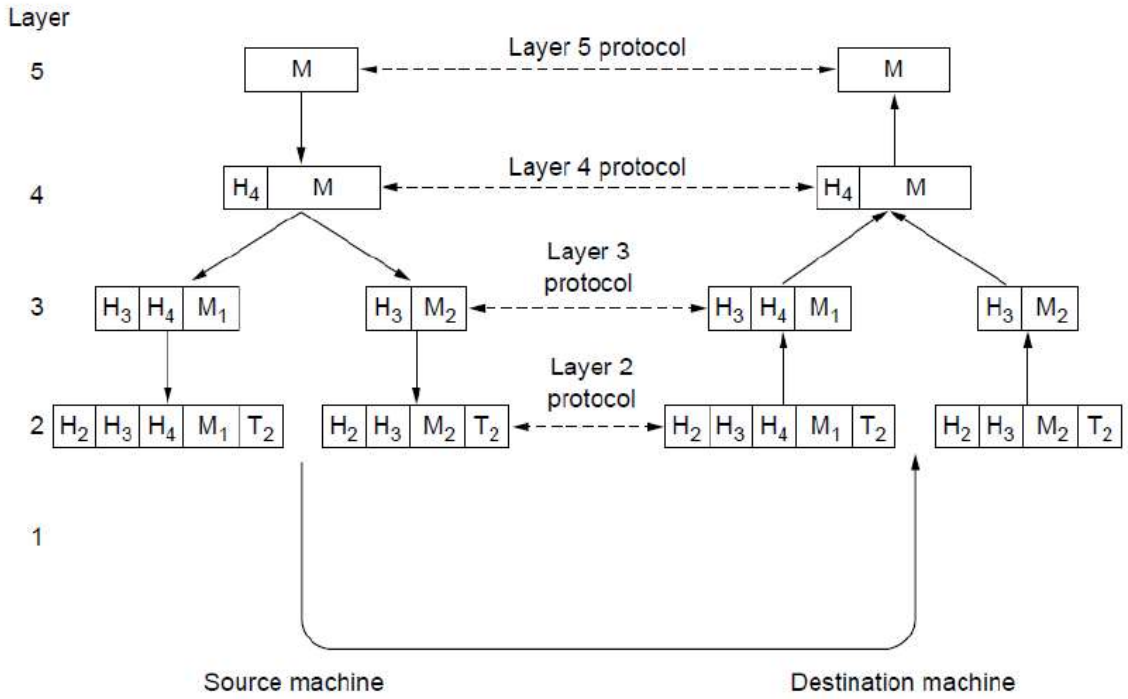
Protocol Layers (2)

- Example: the philosopher-translator-secretary architecture
- Each protocol at different layers serves a different purpose



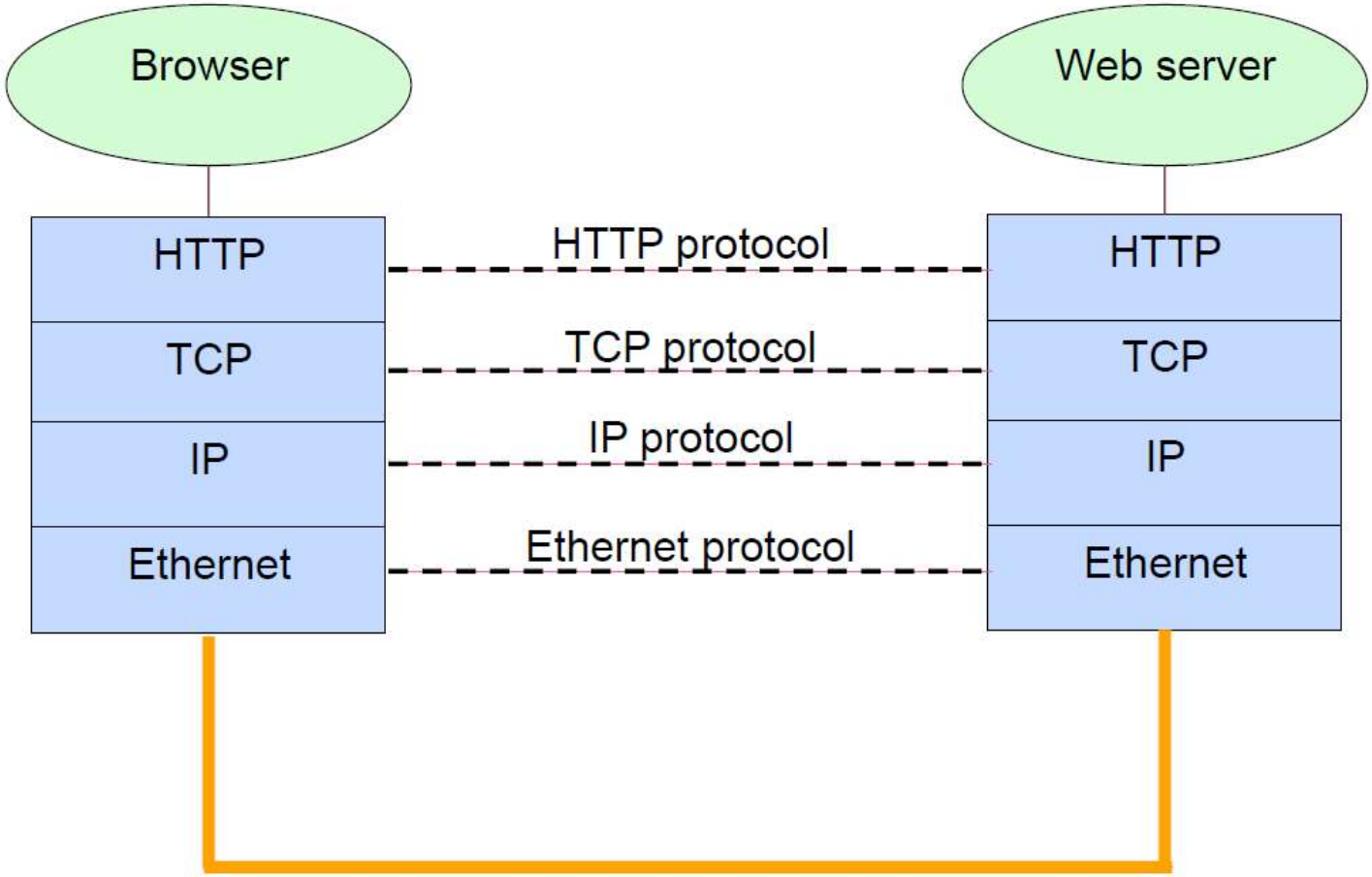
Protocol Layers (3)

Each lower layer adds its own header (with control information) to the message to transmit and removes it on receive

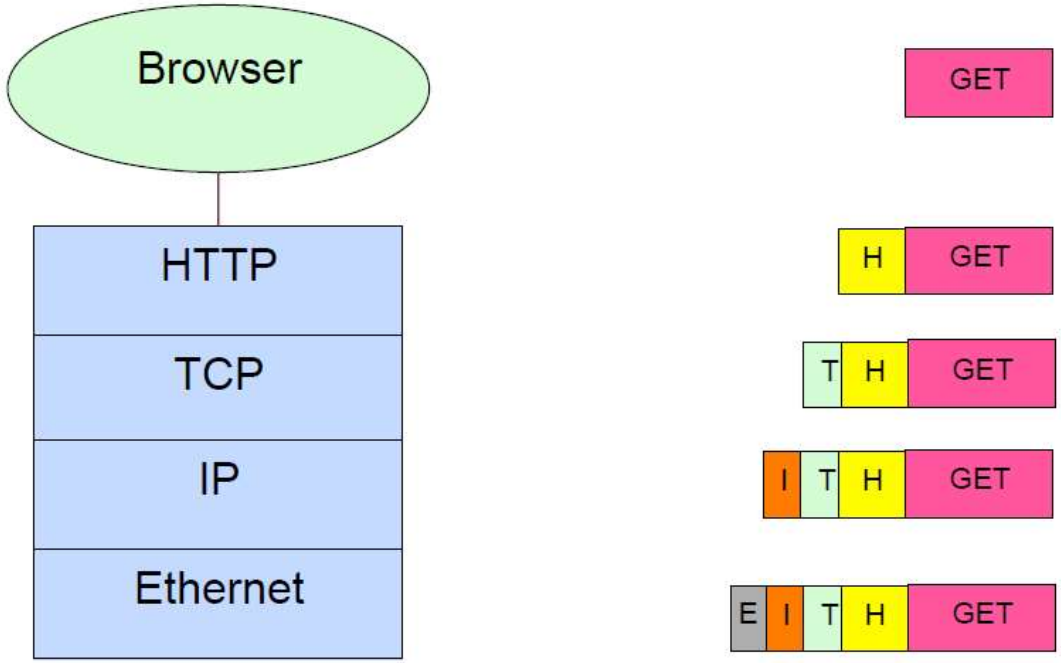


Layers may also split and join messages, etc.

Example Protocol Stack



Encapsulation



Design Issues for the Layers

Each layer solves a particular problem but must include mechanisms to address a set of recurring design issues

Issue	**Example mechanisms at different layers**
Reliability despite failures	Codes for error detection/correction Routing around failures
Network growth and evolution	Addressing and naming Protocol layering
Allocation of resources like bandwidth	Multiple access Congestion control
Security against various threats	Confidentiality of messages Authentication of communicating parties

Connection-Oriented vs. Connectionless

Service provided by a layer may be kinds of either:

- Connection-oriented, must be set up for ongoing use (and torn down after use), e.g., phone call
- Connectionless, messages are handled separately, e.g., postal delivery

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

Service Primitives (1)

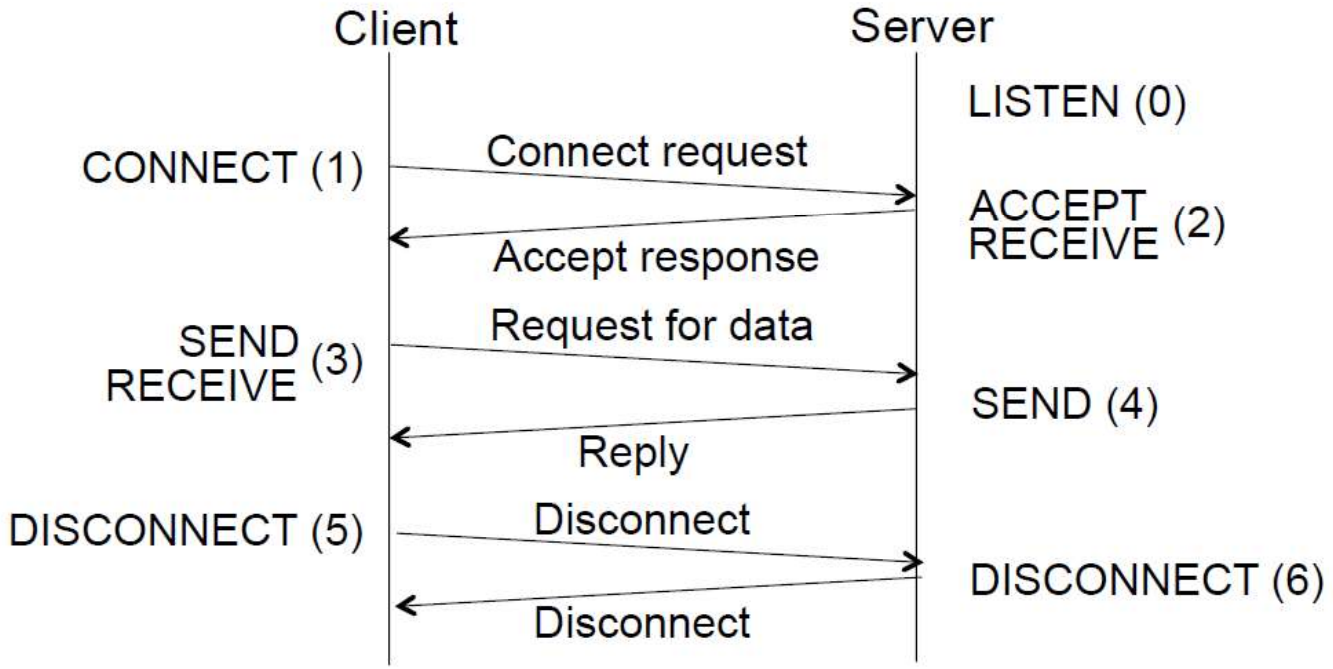
A service is provided to the layer above as primitives

Hypothetical example of service primitives that may provide a reliable byte stream (connection-oriented) service:

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Service Primitives (2)

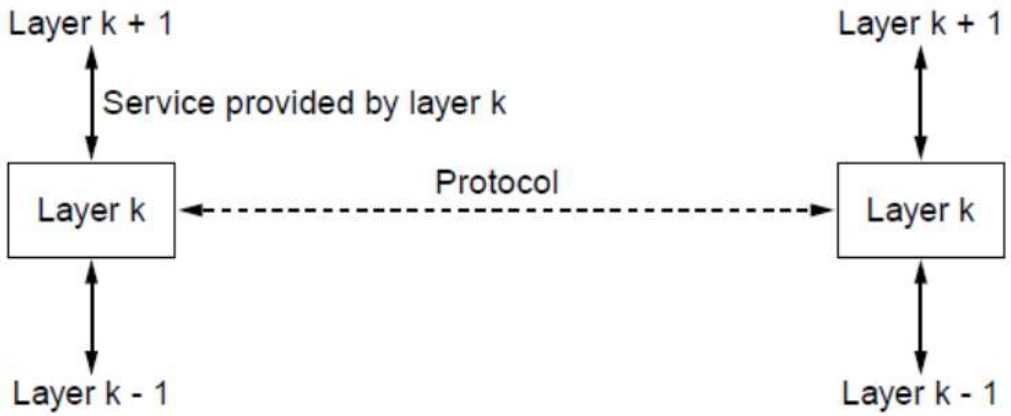
Hypothetical example of how these primitives may be used for a client-server interaction



Relationship of Services to Protocols

Recap:

- A layer provides a service to the one above (vertical)
- A layer talks to its peer using a protocol (horizontal)

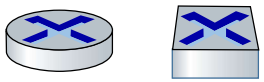


The Internet: overview



Billions of connected computing *devices*:

- *hosts = end systems*
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers*, *switches*

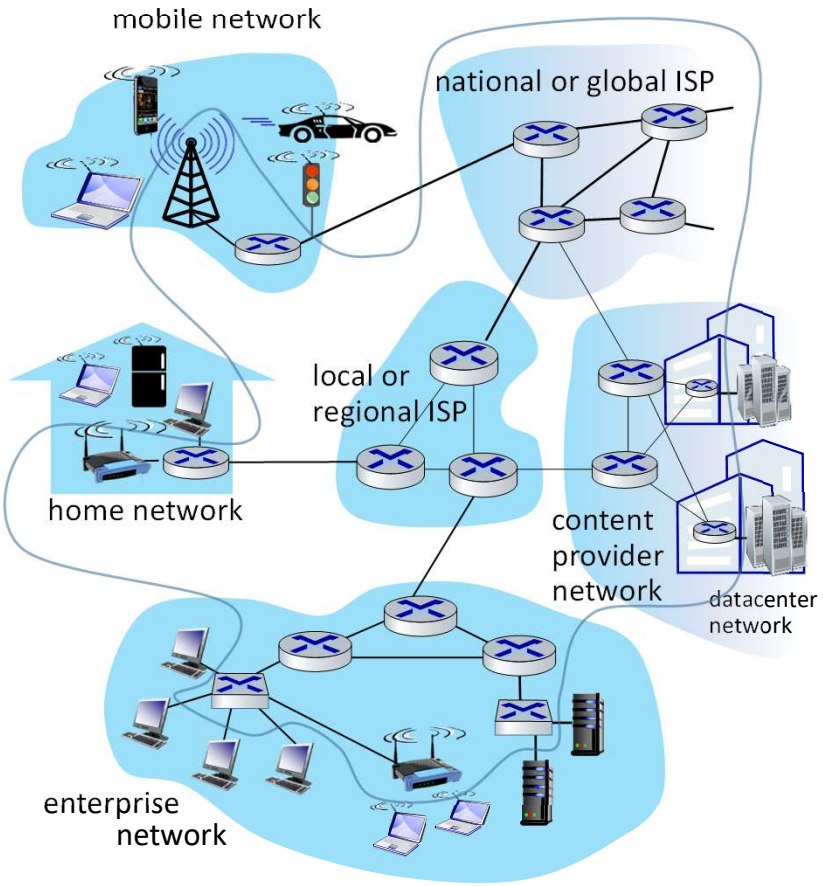


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

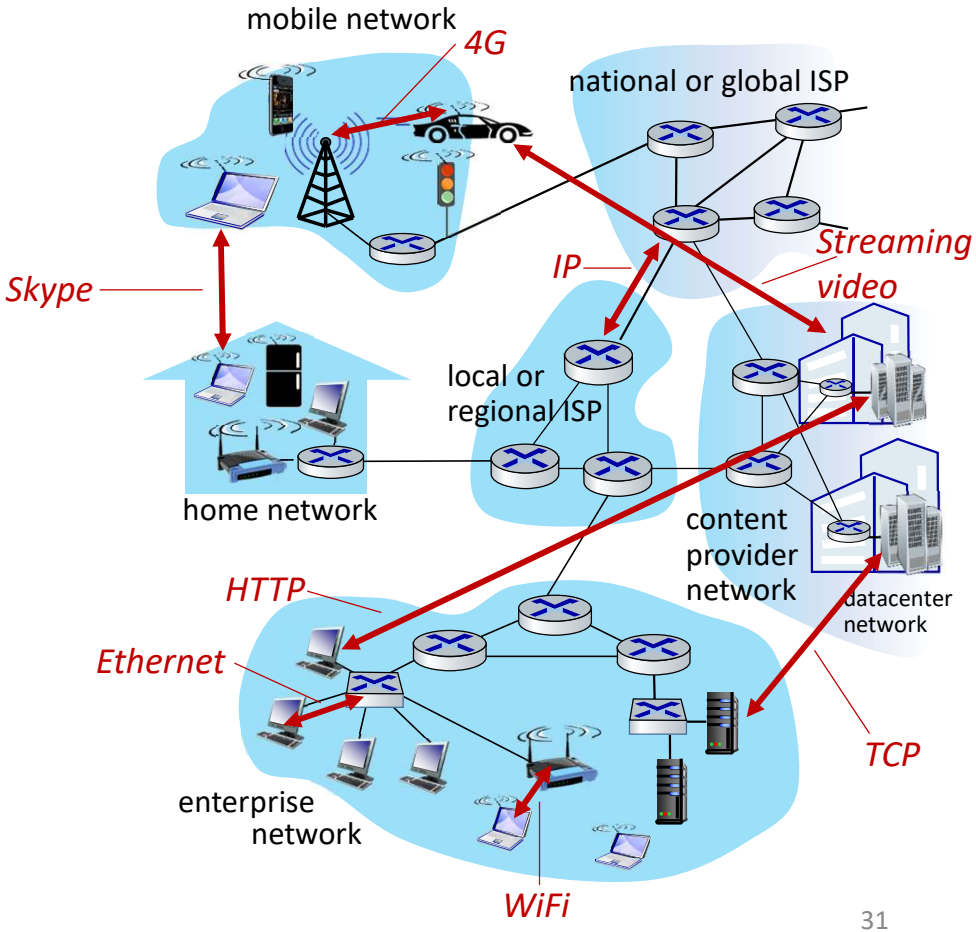
Networks

- collection of devices, routers, links: managed by an organization



The Internet: overview

- *Internet:* "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Zoom, TCP, IP, WiFi, 4/5G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



Reference Models

Reference models describe the layers in a network architecture

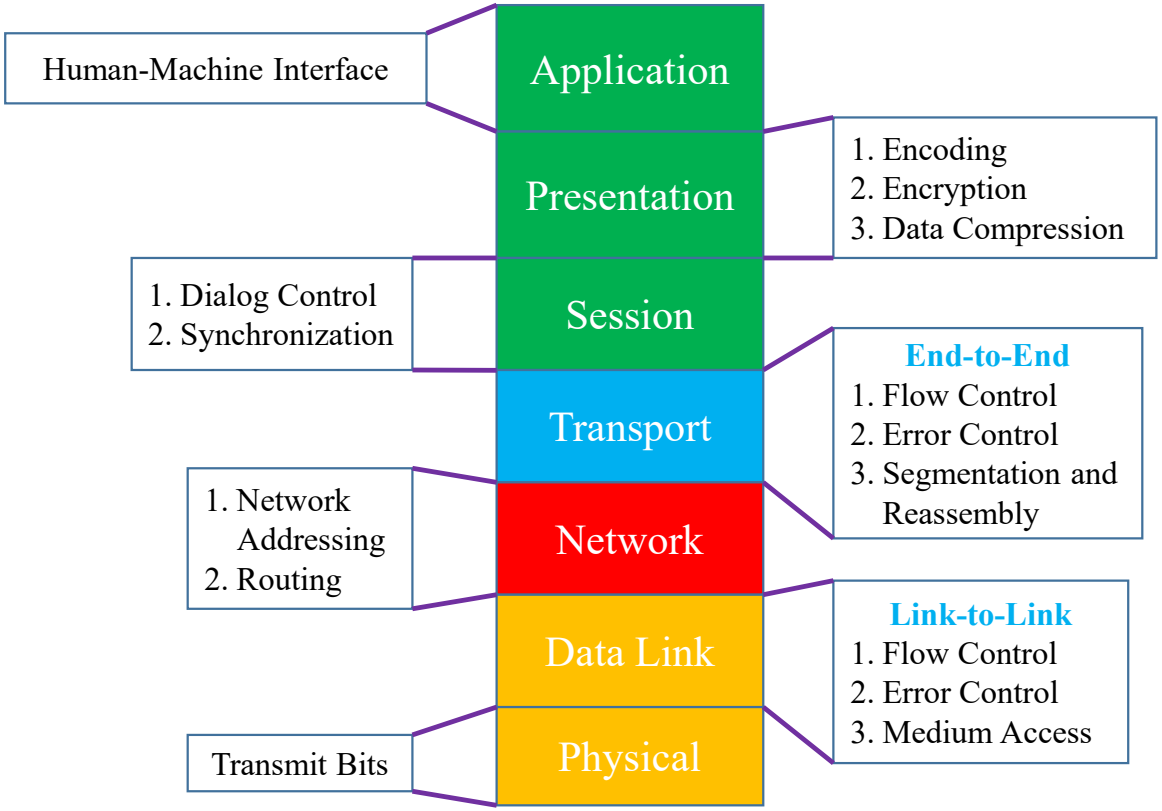
- OSI reference model
- TCP/IP reference model
- Model used for this text
- Critique of OSI models and protocols
- Critique of TCP/IP model

The OSI Reference Model (1)

Principles for the seven layers

- Layers created for different abstractions
- Each layer performs well-defined function
- Function of layer chosen with standards in mind
- Minimize information flow across layer interfaces
- Find the optimum number of layers

OSI Layers & Functionalities





Sender

OSI Model

Receiver



Sender	Receiver
Application Layer	Application Layer
Presentation Layer	Presentation Layer
Session Layer	Session Layer
Transport Layer	Transport Layer
Network Layer	Network Layer
Datalink Layer	Datalink Layer
Physical Layer	Physical Layer

Relationship of OSI Layers

- Each layer of the OSI model must communicate with the layer above and below it
  - For example, the Presentation layer must communicate with the Application layer (one above) and the Session layer (one below)

- As data passes down through the OSI layers, each layer (except Physical) adds some information to the data

Relationship of OSI Layers

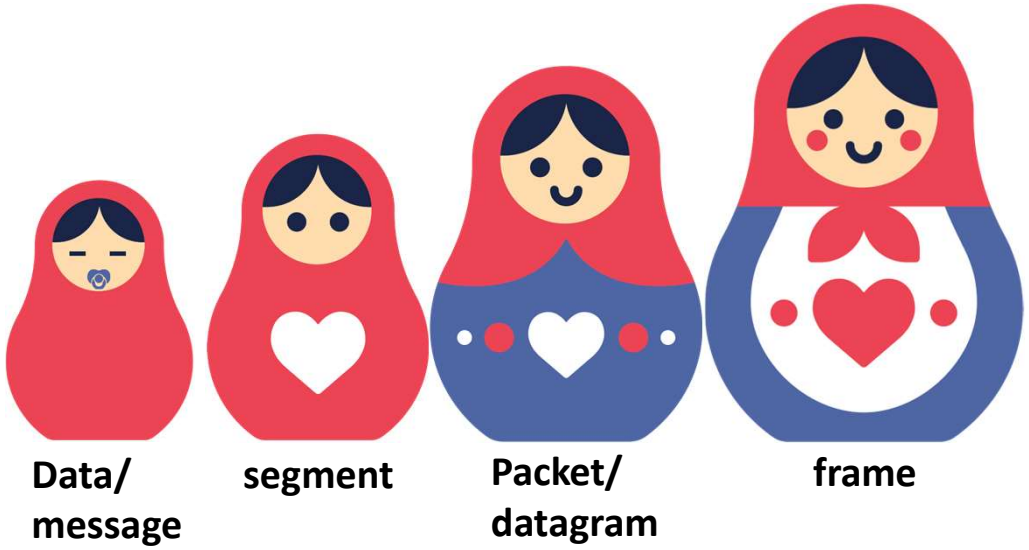
- When data reaches the receiving computer, the information added by each layer of the OSI model is read and processed by the corresponding layer on the receiving computer

- This is referred to as peer-layer communications

Encapsulation

Matryoshka dolls (stacking dolls)

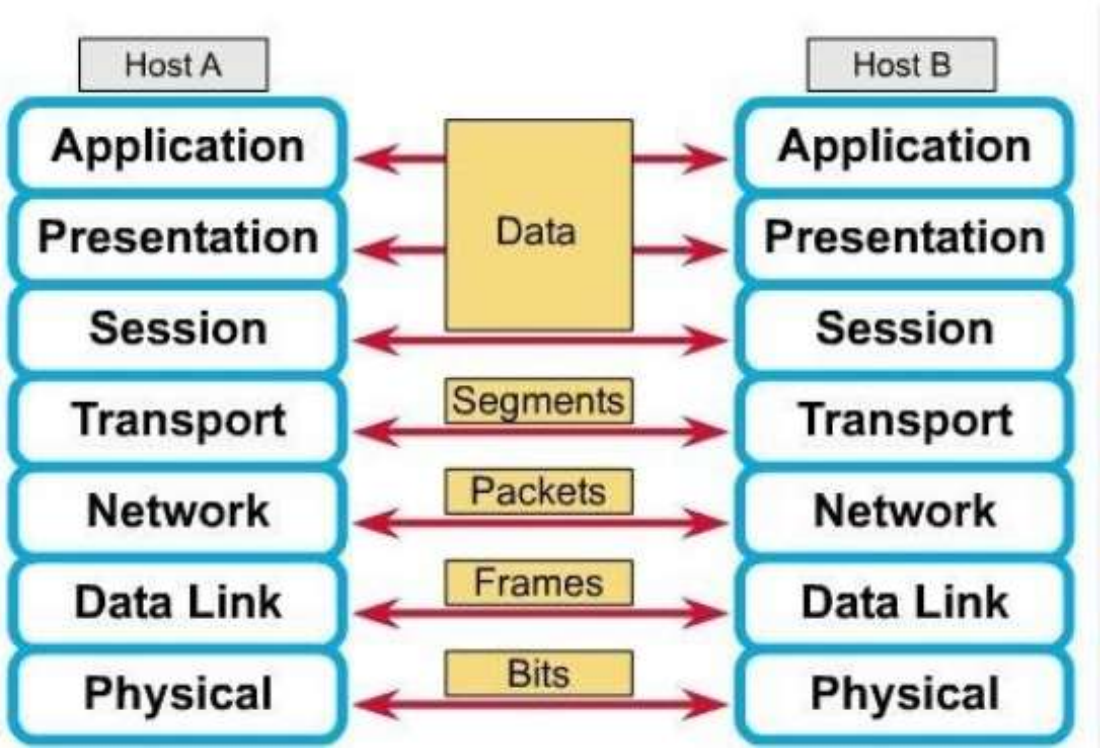


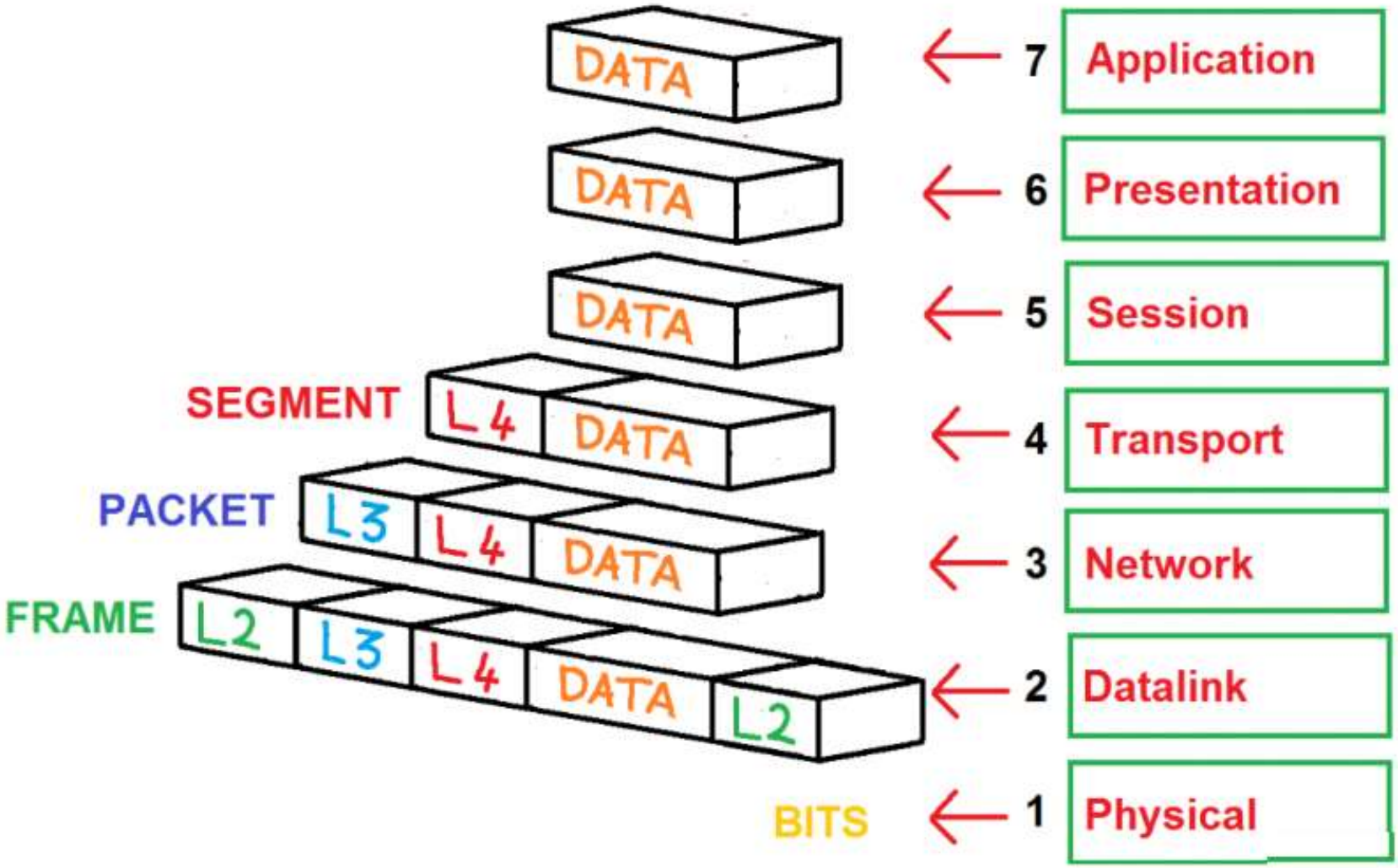


Credit: https://dribbble.com/shots/7182188-Babushka-Boi

OSI

- A Protocol Data Unit or PDU refers to the entity created at each layer of the OSI Model or the TCP-IP model.




DATA
DATA
DATA
SEGMENT
L4
DATA
PACKET
L3
L4
DATA
FRAME
L2
L3
L4
DATA
L2
BITS
7
Application
6
Presentation
5
Session
4
Transport
3
Network
2
Datalink
1
Physical

OSI Layer 1 – Physical



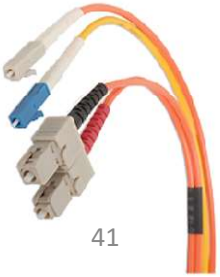
- transfer of bits — the 1's and 0's
- represents the physical medium which is carrying the traffic between two nodes.
- Ethernet -electric pulses.
- Wifi - radio waves.
- Fiber - pulses of light.



- Aside from the physical cable, Repeaters and Hubs also fall in this layer.
  - Repeaters
    - Restores the strength of an attenuated signal.
    - Used to increase the transmission distance.
    - Does not filter data traffic.



  - Hubs
    - Multi-port repeater.
    - Interconnects several computers.
    - Does not filter data traffic.



OSI L1 – Physical

- Responsible for transmitting bits from one computer to another
- Physical topology
- Network connection types
  - Number of pins on the network connector
  - Passive and active hubs, cables, repeaters, transceivers, etc.
- Defines the electrical details
  - Bit encoding—what represents 0 and 1
  - How data will be synchronized
  - Baseband/broadband transmission

OSI L 2 – Data Link



- Transmitting n receiving 0's and 1's from the wire.
- Network Interface Card (NIC)
- 1's and 0's are made into chunks - Frames.
- Media Access Control address, or MAC address.
  - uniquely identifies each individual NIC.
- Each NIC is pre-configured with a MAC address by the manufacturer; (Burned In Address (BIA)).
- Switch also operates at this layer.
- hop to hop.

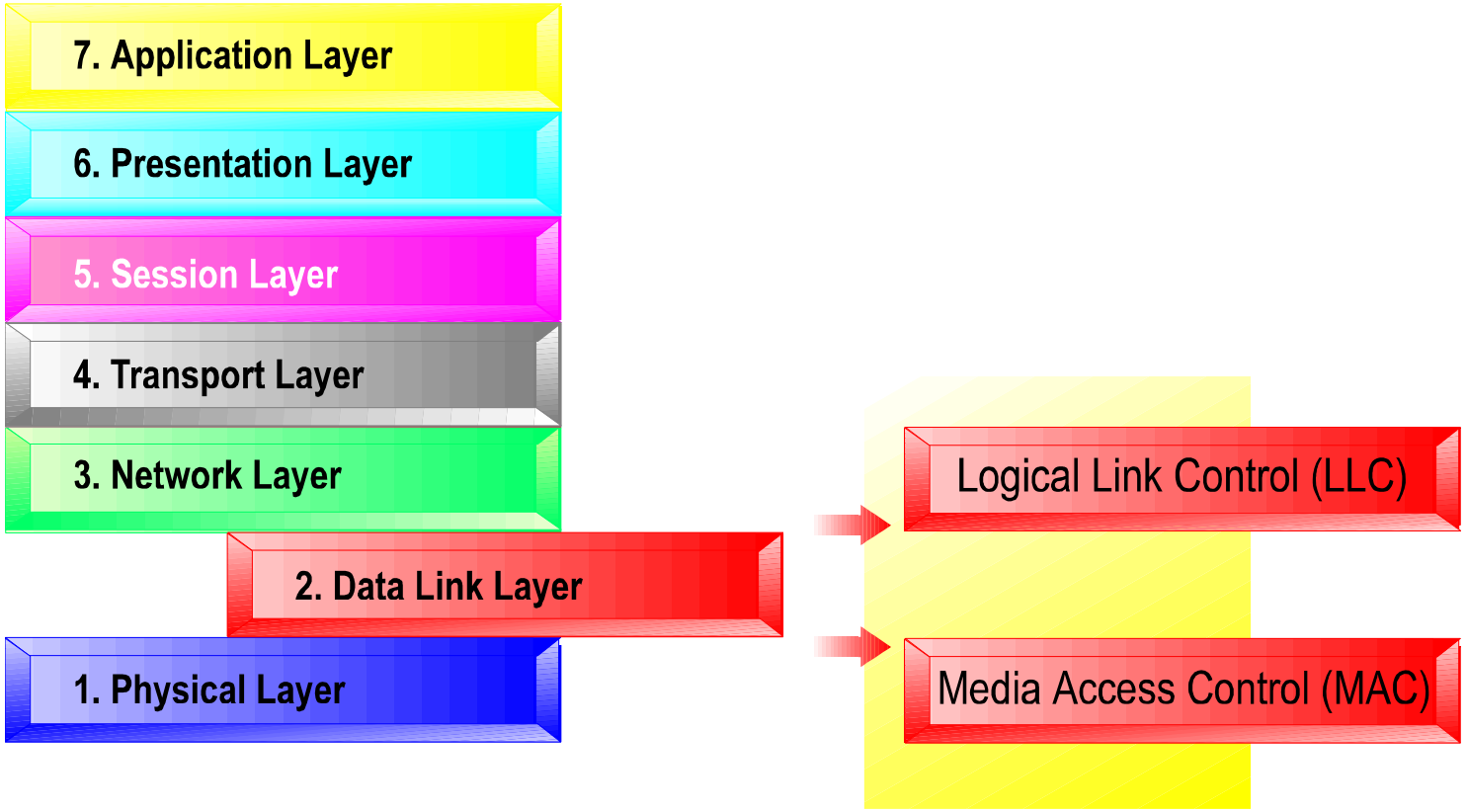
OSI L2 – Data Link Layer

- Accepts packets of data from Network layer and packages the data into data units called frames
- Adds information such as frame type and physical address
- Responsible for providing error-free transfer of data frames
  - Frame Check Sequence—error checking algorithm is added at the end of each frame so that receiving computer can detect damaged frames and request that frames be resent
    - Cyclic Redundancy Check (CRC)—most commonly used algorithm

OSI L2 – Data Link Layer

- Intelligent hubs, bridges, and switches are addressed by the Data Link Layer
- Defines how media will be accessed
- Divided by 802 model into two sub-layers:
  - Logical Link Control (LLC)
  - Media Access Control (MAC)

Project 802 Logical Link Control and Media Access Control sub-layers



OSI L2 – Data Link Layer

- Logical Link Control
  - Uses error check algorithm (FCS) to verify that data is received correctly
  - Ensures that the rate of transmission is appropriate for the receiving computer
  - Provides the link between the Data Link layer and the Network layer

OSI L2 – Data Link Layer

- Media Access Control
  - Controls the way multiple devices share the same media channel
    - Defines the method the NAC will use to determine when to send data
  - Communicates directly with the network adapter card
  - MAC (physical) address for network adapter cards provide unique identifier for each NAC

OSI L 3 – Network

- **packet delivery from end to end**.
- Logical addressing scheme of every node on the internet
  - Internet Protocol address, or the IP Address.
  - Logical addressing scheme of every node on the internet
  - not a permanent identification of a computer.

- Routers are Network Devices that operate at Layer 3 of the OSI model.

OSI Model – Layer 2 vs. Layer 3

- Need for MAC when we already have IP address??

- Both addressing schemes accomplish different functions:

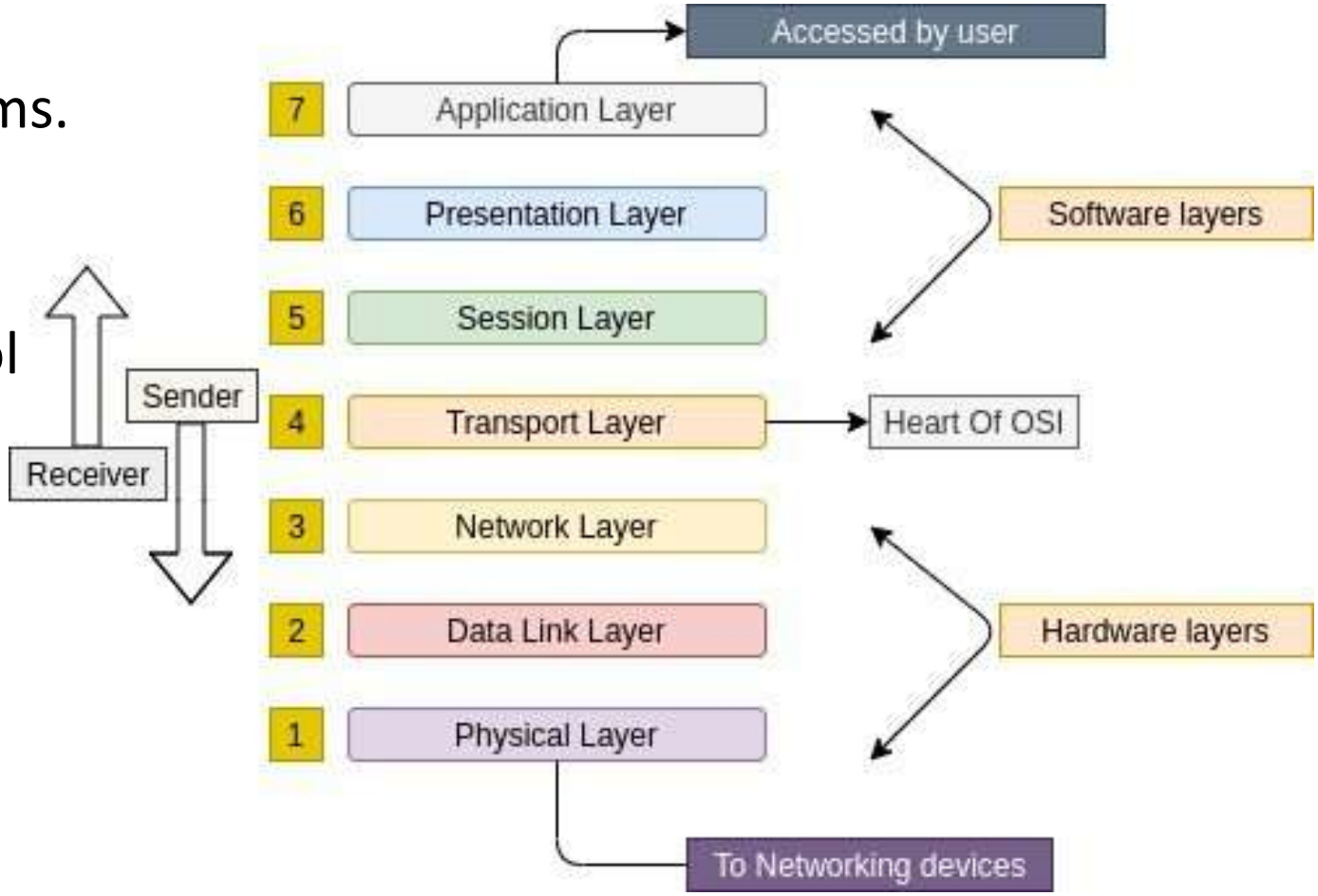
- Layer 2 MAC addresses - packet delivery from hop to hop.
- Layer 3 IP addresses - packet delivery from end to end.

OSI L3 – Network Layer

- Responsible for addressing messages across networks
- Translates network logical addresses into physical machine addresses
- Breaks large segments into smaller packets
- Determines routing across networks
- Supports communications between logically separate networks
  - Routers and gateways (devices that pass data from one network to another) operate in the Network layer

OSI L4 – Transport

- distinguishing network streams.
- addressing scheme known as Port Numbers.
- Transmission Control Protocol (TCP), or the User Datagram Protocol (UDP).
- 65,536 port numbers (each)



OSI-L4 - Transport Layer

- Ensures data is delivered error-free in sequence and without duplication or loss
- Breaks large packets from the Session layer into segments to be sent to the receiving computer
- Adds sequencing information to segments
- Reassembles the segments into messages
- Sends acknowledgement to sending computer

OSI L5-Session Layer

- Responsible for establishing and maintaining communications between two nodes on the network
- Allows applications on separate computers to share a connection (called a session)
- Provides data synchronization and checkpointing so that if there is a network failure only the data sent after the checkpoint must be resent
- Provides name recognition and security needed to allow two applications to communicate

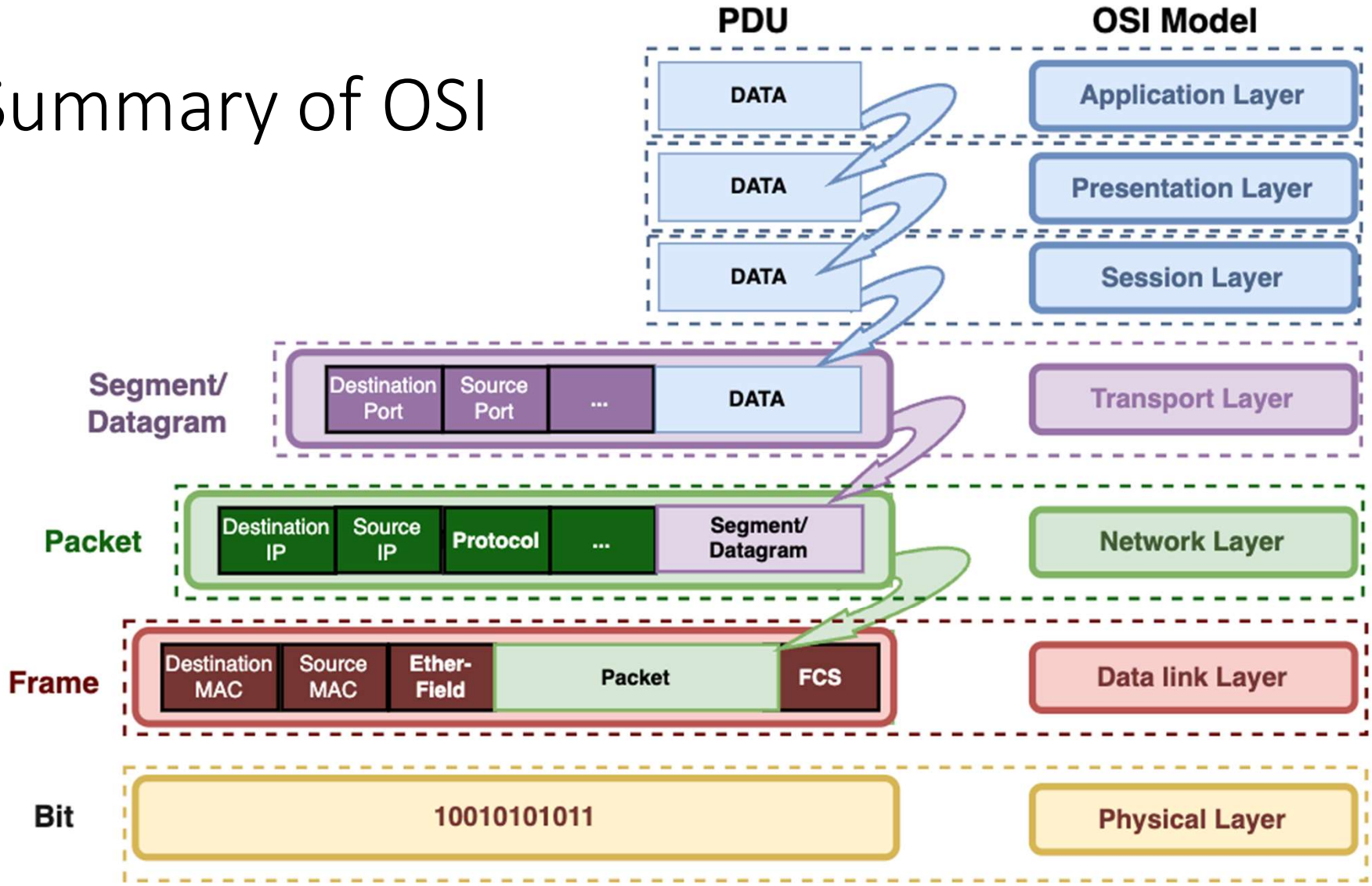
OSI L6- Presentation Layer

- Translates data between the format the network requires and the format the application expects
- Responsible for encrypting data, changing or converting the character set and interpreting graphic commands
- Manages data compression to reduce number of bits to be transmitted

OSI L7- Application Layer

- Provides services to support user applications, such as file transfer, database access, and e-mail
  - Not a particular application—but services that are provided to applications

Summary of OSI



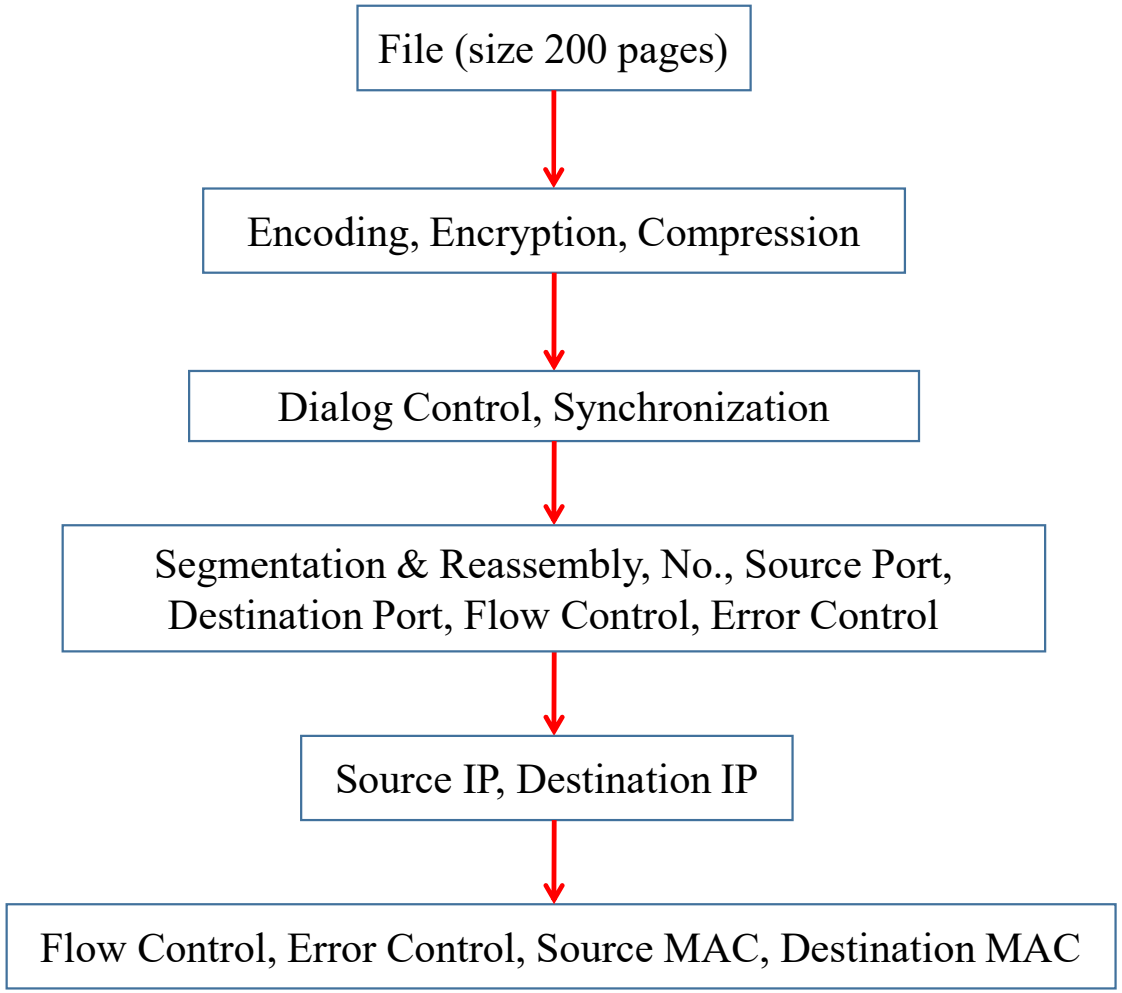
Addressing

- MAC Address—physical address burned onto NIC card
  - Unique address for each NIC card produced in the world
  - Consists of a Block ID and a Device ID
    - Each manufacturer has one or more Block IDs
  - Added to frame by Data Link Layer—MAC sublayer

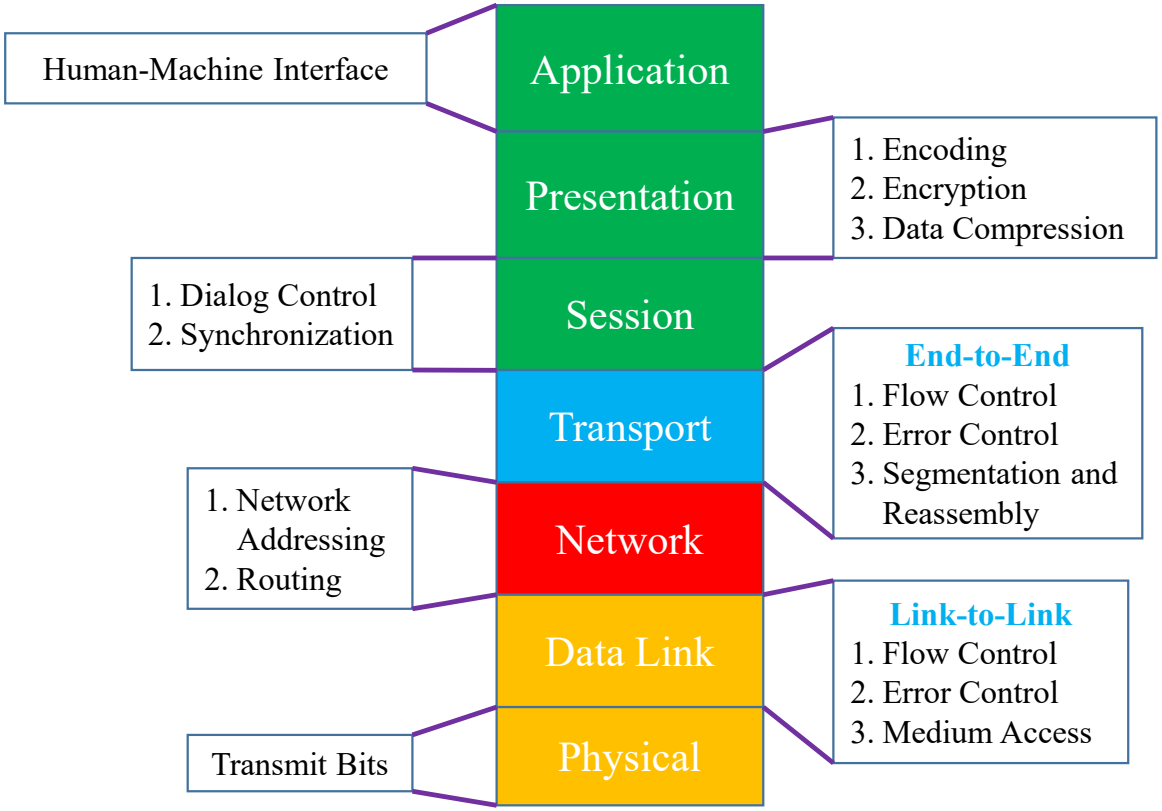
Addressing

- Network address—logical address assigned to a network device that identifies the network that a device belongs
  - Can be assigned automatically when a computer is turned on
    - DHCP server provides IP (network) addresses to computers
  - Can be assigned manually so that address remains the same each time computer is turned on
    - Addresses for servers and printers are assigned manually so that other devices can always locate them

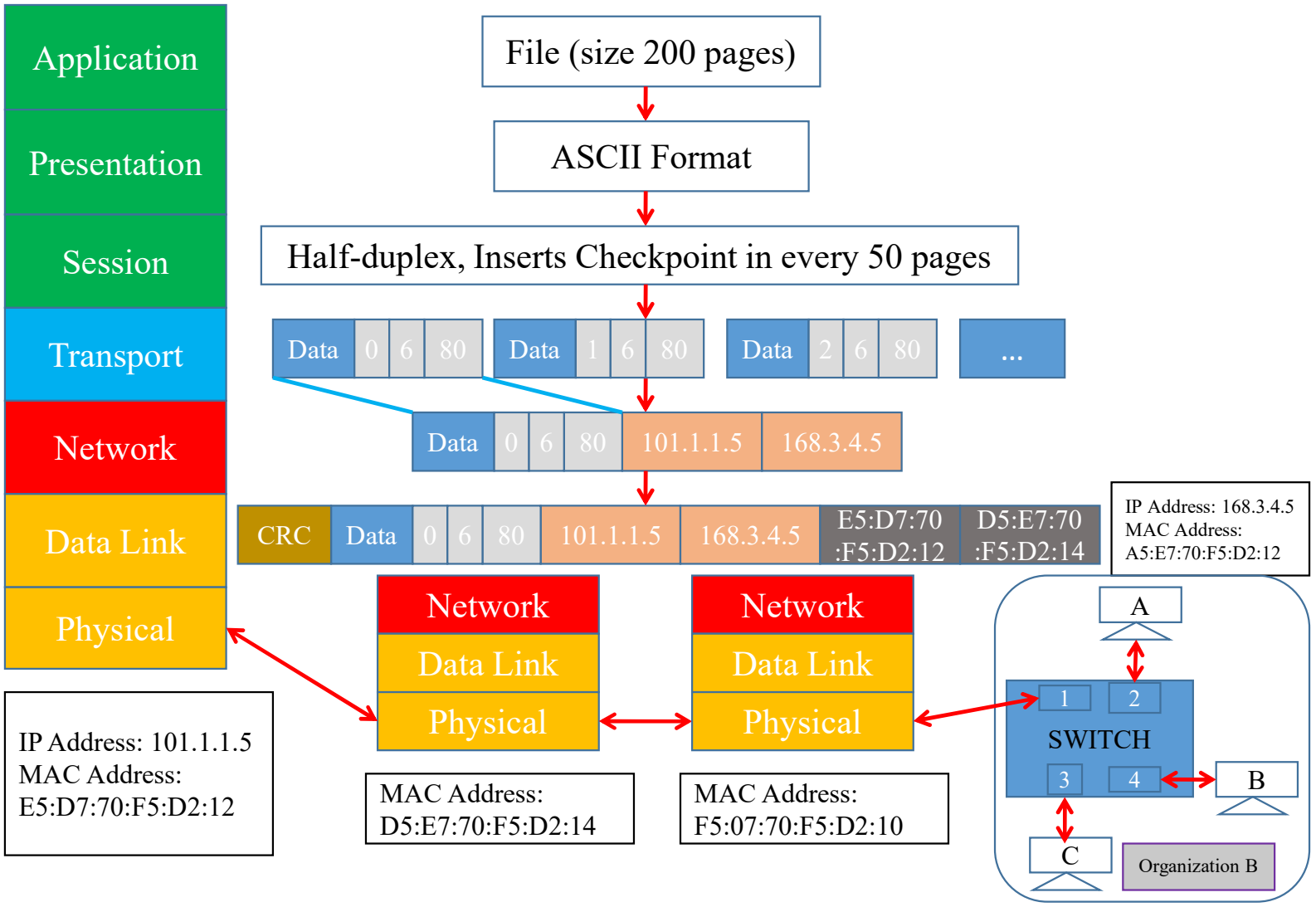
Requirements



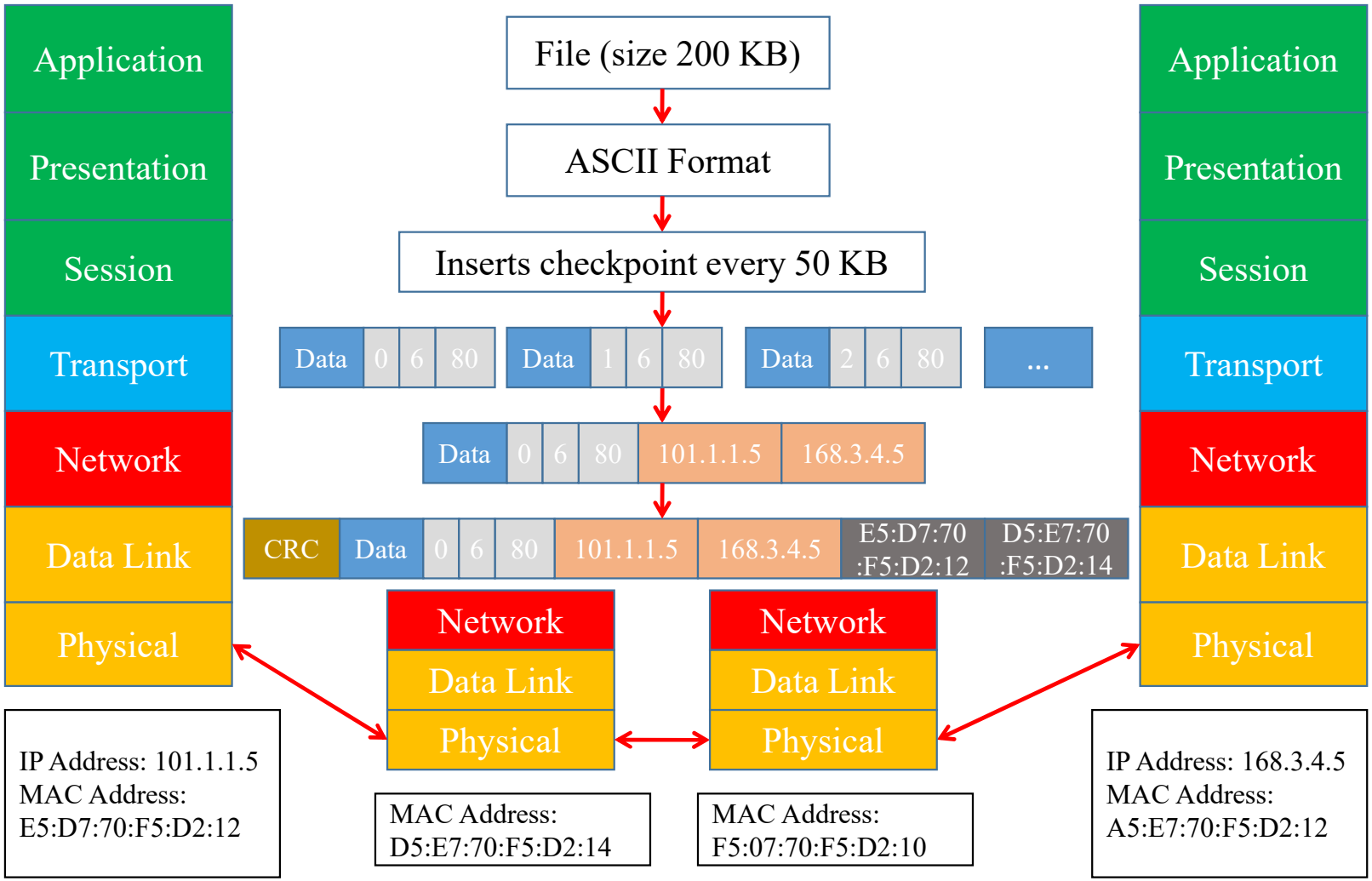
OSI Layers & Functionalities



Example

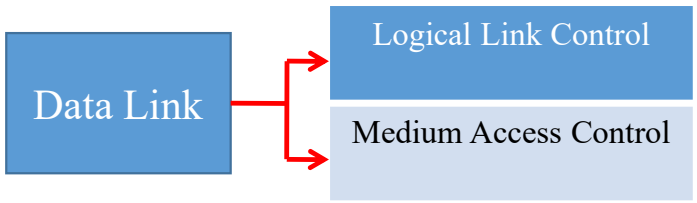

Application
Presentation
Session
Transport
Network
Data Link
Physical
File (size 200 pages)
ASCII Format
Half-duplex, Inserts Checkpoint in every 50 pages
Data 0 6 80
Data 1 6 80
Data 2 6 80
...
Data 0 6 80 101.1.1.5 168.3.4.5
CRC Data 0 6 80 101.1.1.5 168.3.4.5 E5:D7:70 :F5:D2:12 D5:E7:70 :F5:D2:14
IP Address: 168.3.4.5
MAC Address:
A5:E7:70:F5:D2:12
IP Address: 101.1.1.5
MAC Address:
E5:D7:70:F5:D2:12
Network
Data Link
Physical
MAC Address:
D5:E7:70:F5:D2:14
Network
Data Link
Physical
MAC Address:
F5:07:70:F5:D2:10
A
1
2
SWITCH
3
4
B
C
Organization B

Example (Contd.)

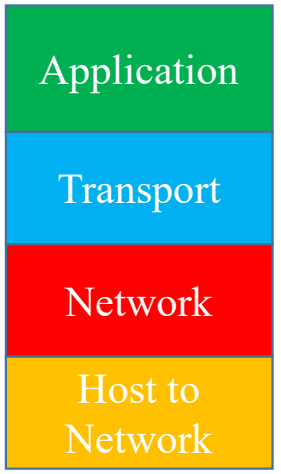


Modifications

Data Link Layer



TCP/IP Protocol Suite

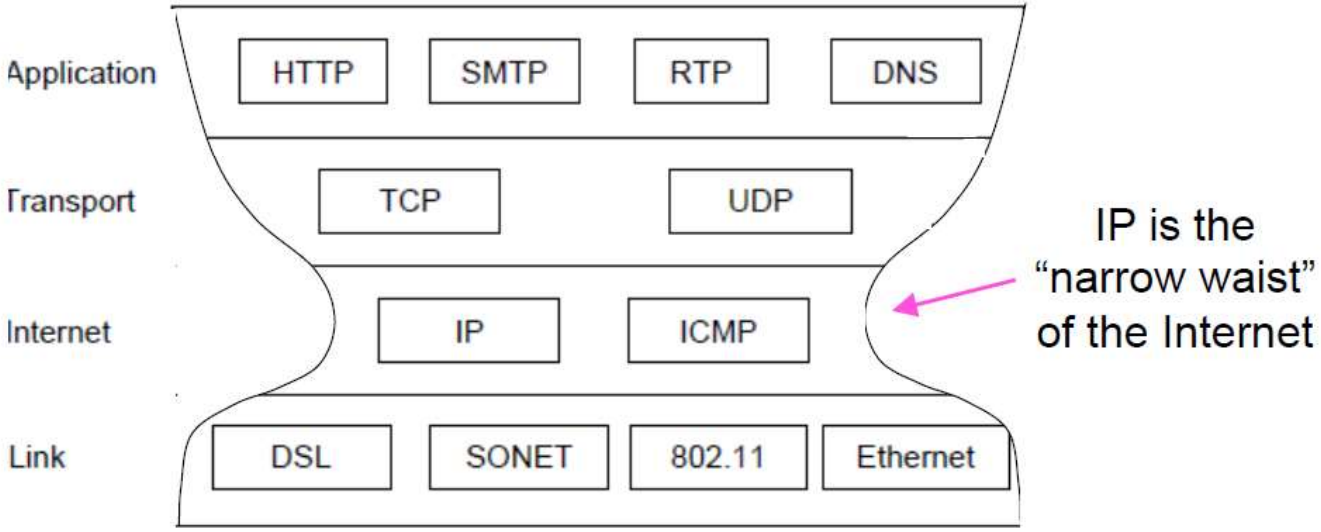


The TCP/IP Reference Model Layers

- Link layer
- Internet layer
- Transport layer
- Application layer

TCP/IP Reference Model

A four layer model derived from experimentation; omits some OSI layers and uses the IP as the network layer.

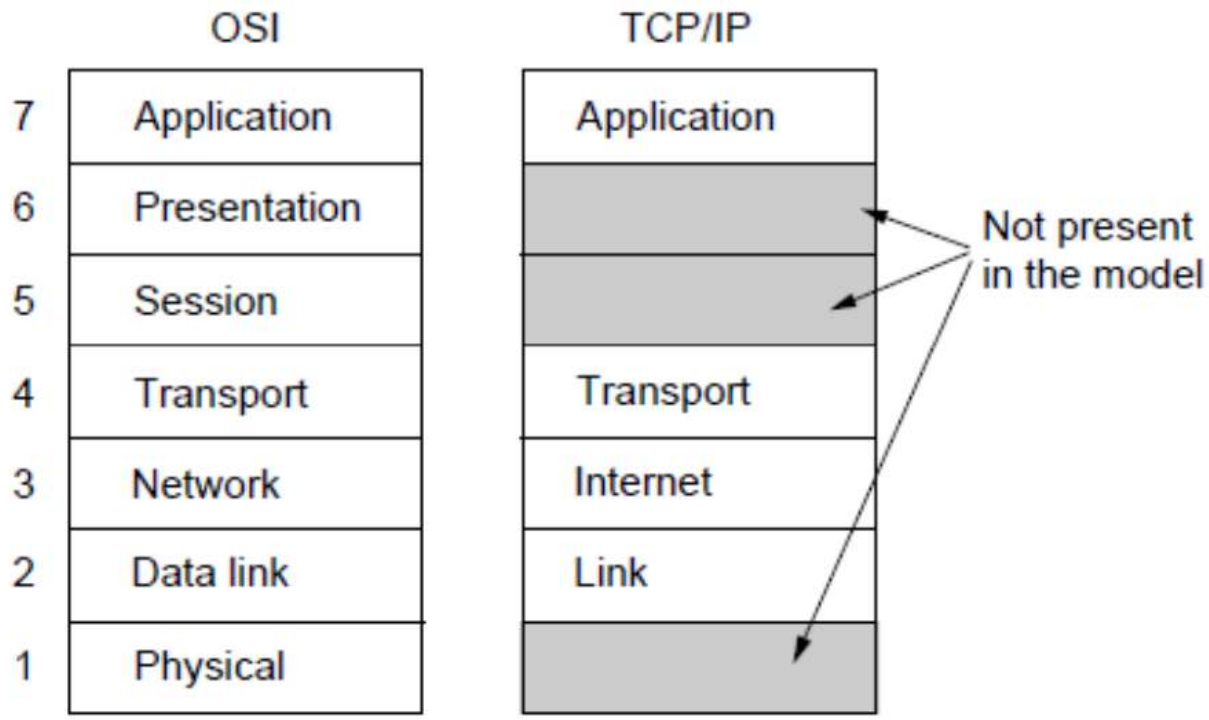


Protocols are shown in their respective layers

Other Internet Protocols

- ARP – Find out which machine has a given IP address
- RARP – Map Ethernet address onto IP address
- DHCP – Dynamically assigns an IP address to a node
- PPP – Used to transmit packets over phone lines
- IGMP – Multicast protocol
- SSH – Allows logins to remote machines
- FTP – Get a file from a remote machine
- NTP – Network time protocol
- SNMP – Management protocol for monitoring devices

The TCP/IP Model vs. OSI



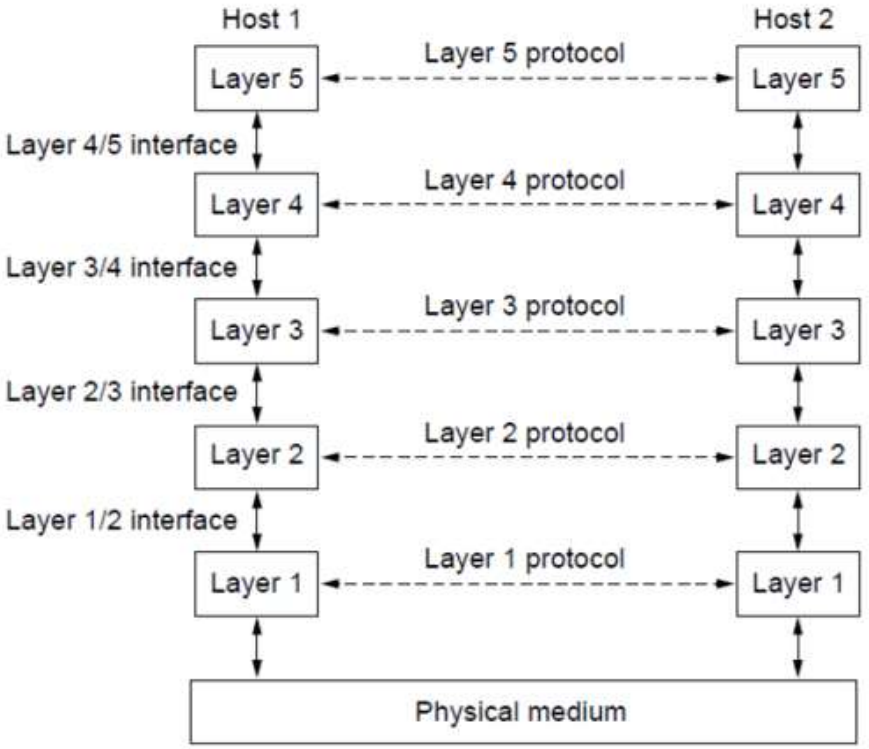
The TCP/IP reference model

OSI Model	PDU	TCP/IP Stack
Application	Data	Application
Presentation		
Session		
Transport	Segment	Transport
Network	Packet	Internet
Data Link	Frame	Network Access/Link
Physical	Bits	

Comparison of the OSI and TCP/IP Reference Models

Concepts central to OSI model

- Services
- Interfaces
- Protocols



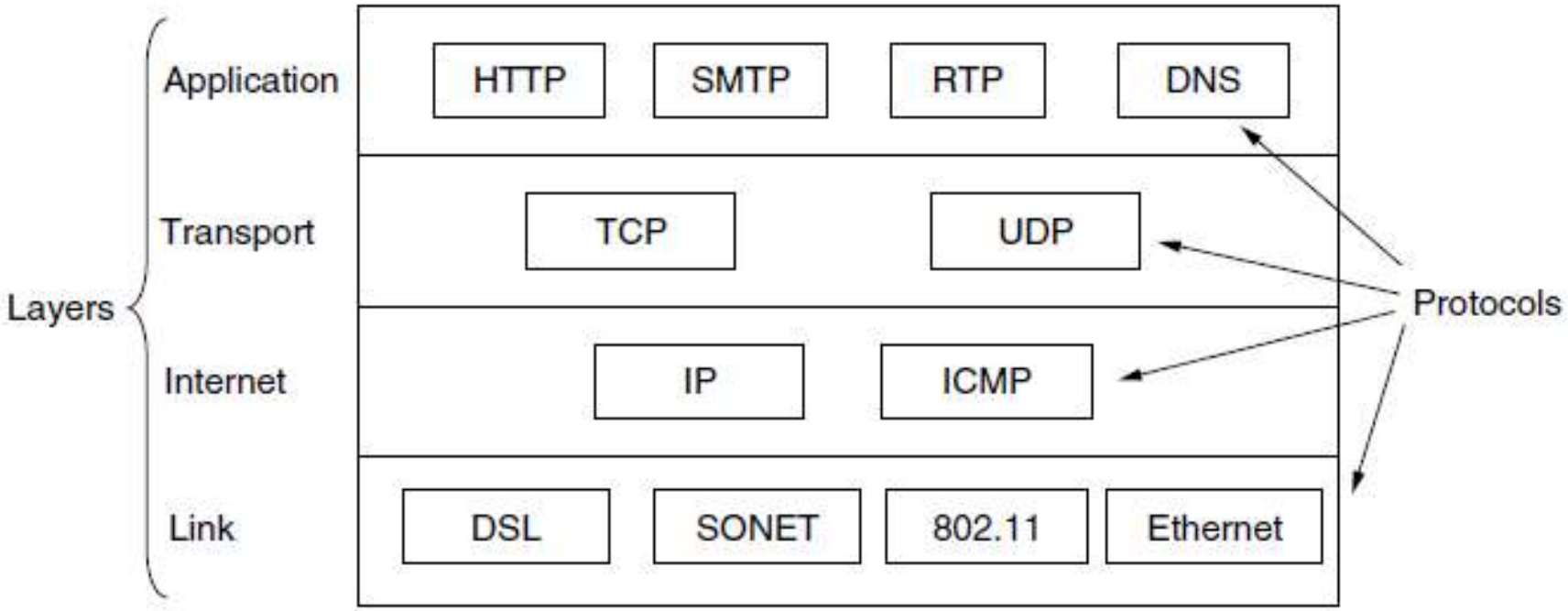


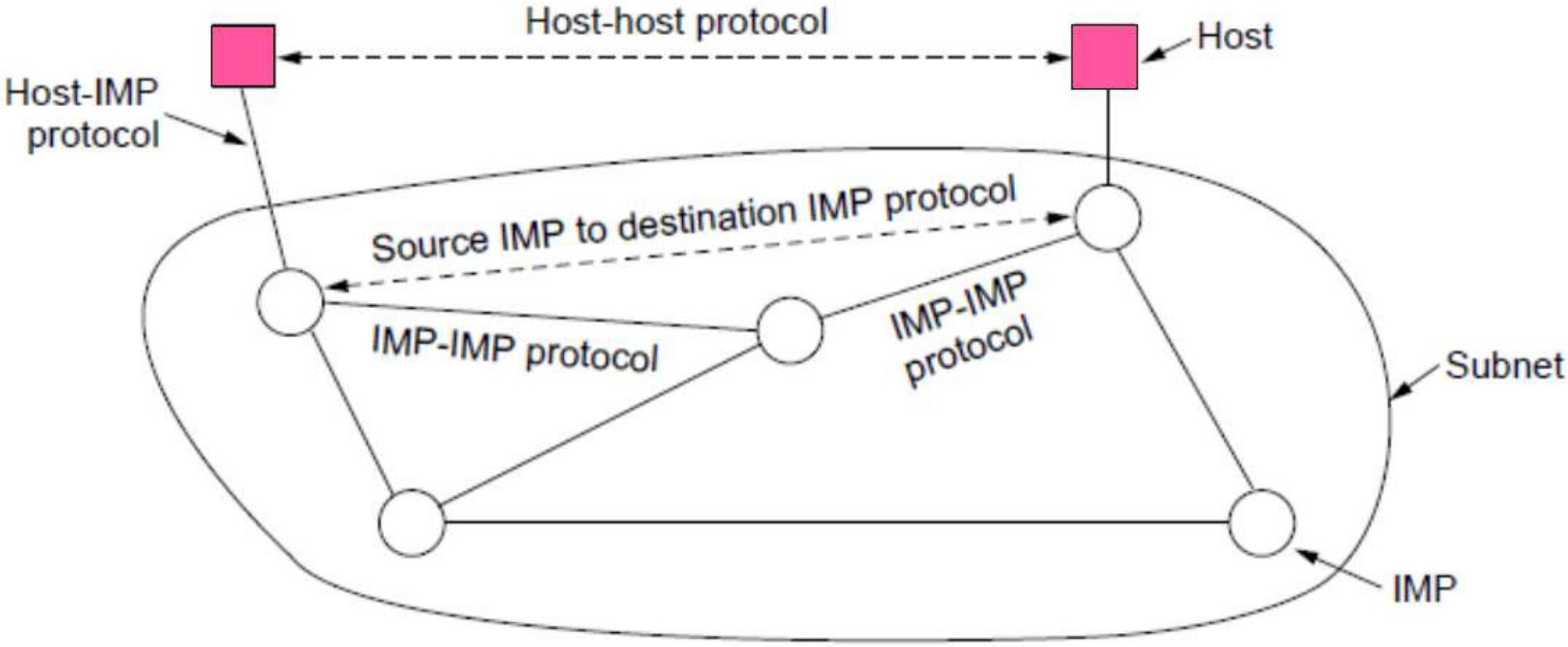
Figure 1-22. The TCP/IP model with some protocols we will study.

Example Networks

- ARPANET
- NSFNET
- Internet
- Third-generation mobile phone networks
- Wireless LANs: 802.11
- RFID and sensor networks

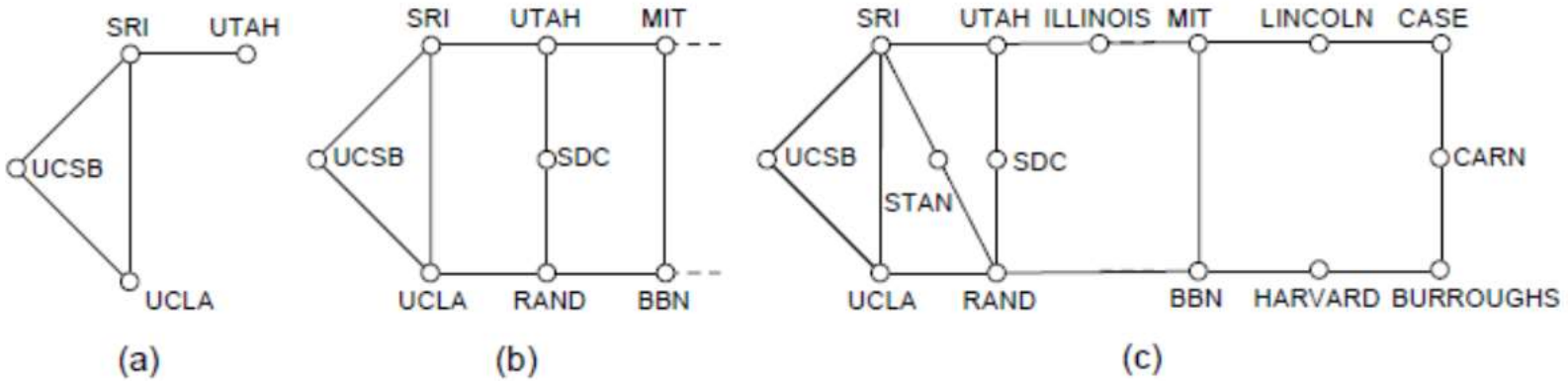
ARPANET

- ARPANET - the First Network
- **ARPANET** – **Advanced Research Projects Agency Network** – the granddad of Internet was a network established by the US Department of Defense (DOD).
- Tt had four nodes at University of California at Los Angeles (UCLA), Stanford Research Institute (SRI), University of California at Santa Barbara (UCSB) and University of Utah. On October 29, 1969, the first message was exchanged between UCLA and SRI.
- The term Internet was coined by Vinton Cerf, Yogen Dalal and Carl Sunshine of Stanford University to describe this network of networks.
- Together they also developed protocols to facilitate information exchange over the Internet.
- Transmission Control Protocol (TCP) still forms the backbone of networking.



The original ARPANET design with 56 kbps lines

ARPANET Early Topology



Growth of the ARPANET.

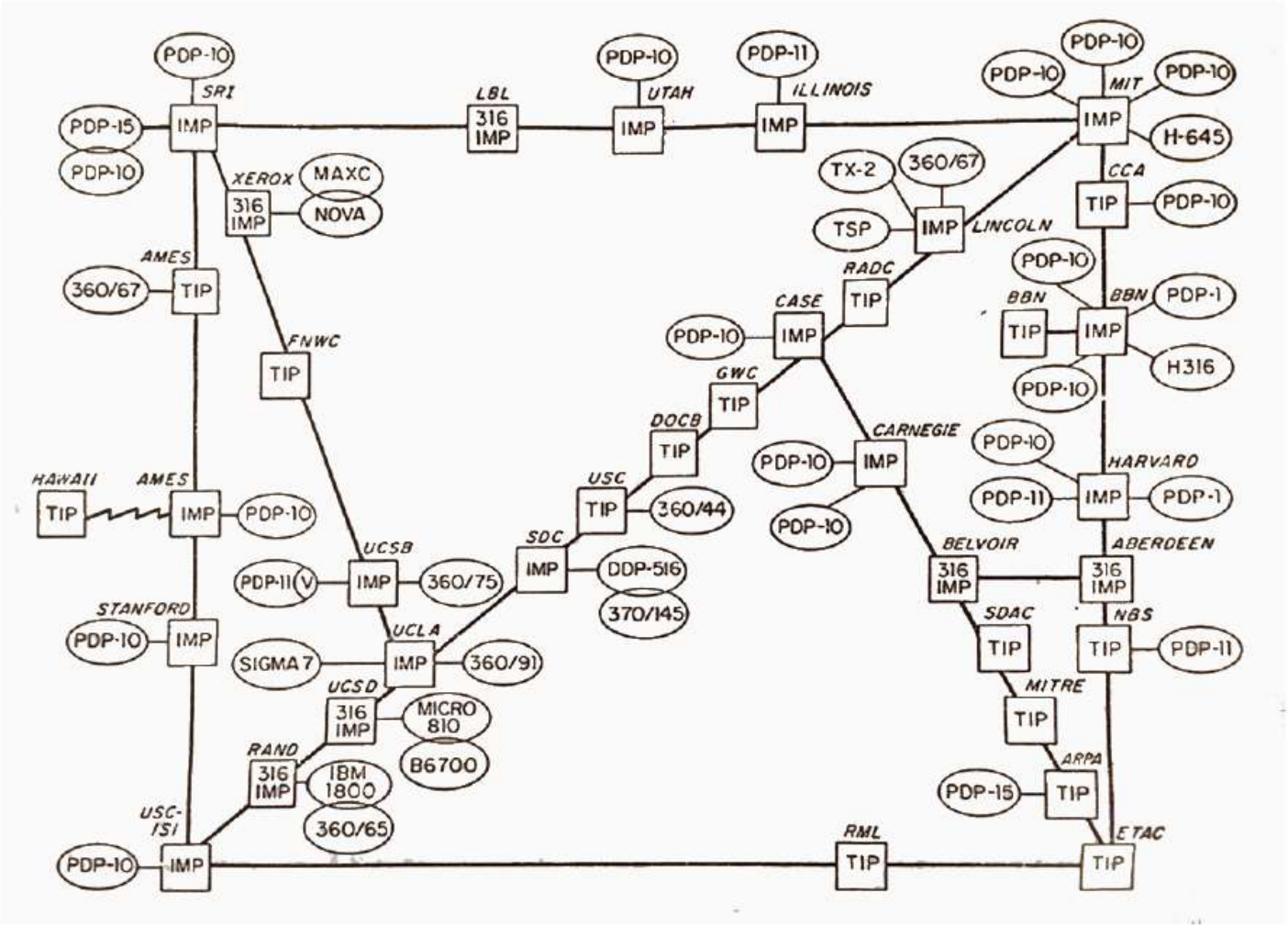
a) December 1969.

b) July 1970.

c) March 1971.

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The ARPANET – 1973 Topology



Host-host protocol used an 8-bit number to identify a host

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Internetworking

It wasn't long before multiple networks got connected:

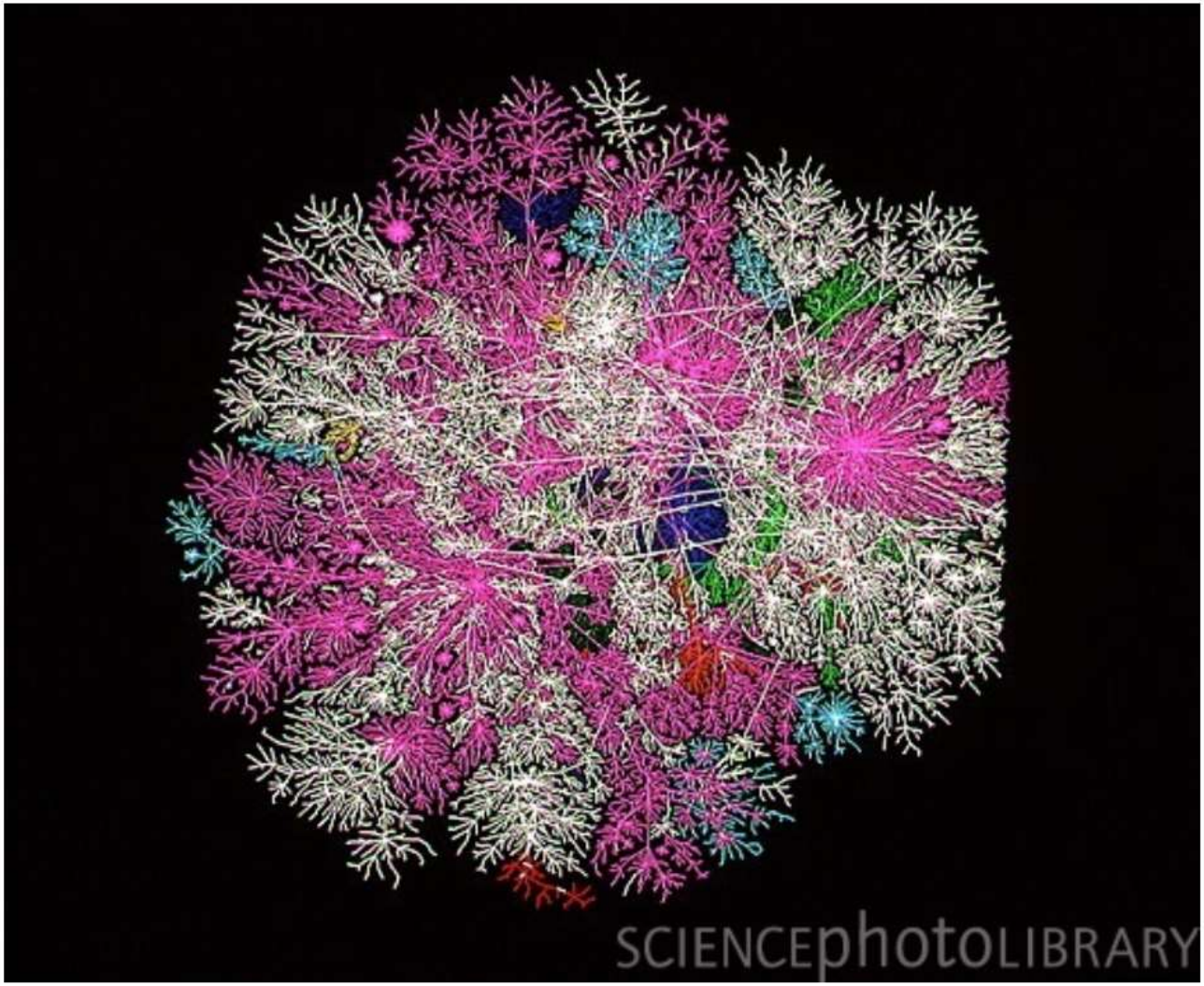
- ARPANET
- NSFNET
- Packet radio net
- Satellite connections to England and Norway



Internet

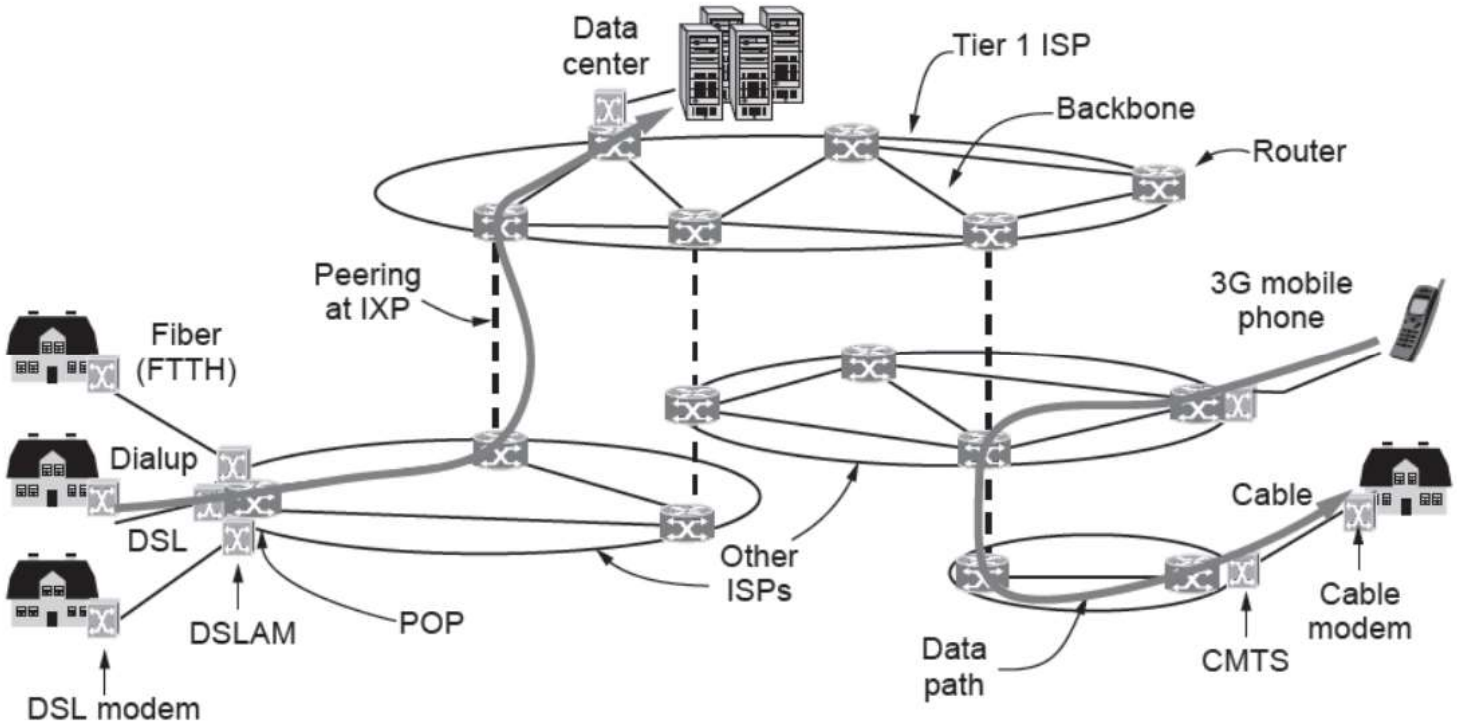
- With commercialization of internet, more and more networks were developed in different part of the world.
- Each network used different protocols for communicating over the network.
- This prevented different networks from connecting together seamlessly.
- In the 1980s, Tim Berners-Lee led a group of Computer scientists at CERN, Switzerland, to create a seamless network of varied networks, called the World Wide Web (WWW).

Visualization of the Internet Now



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Internet (1)



Architecture of the Internet

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Internet (2)

The modern Internet is more complex:

- ISP networks serve as the Internet backbone
- ISPs connect or peer to exchange traffic at IXPs
- Within each network routers switch packets
- Between networks, traffic exchange is set by business agreements
- Customers connect at the edge by many means
  - Cable, DSL, Fiber-to-the-Home, 3G/4G wireless, dialup
- Data centers concentrate many servers ("the cloud")
- Most traffic is content from data centers (esp. video)
- The architecture continues to evolve

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Network Standardization

Standards define what is needed for interoperability

Some of the many standards bodies:

Body	Area	Examples
ITU	Telecommunications	G.992, ADSL H.264, MPEG4
IEEE	Communications	802.3, Ethernet 802.11, WiFi
IETF	Internet	RFC 2616, HTTP/1.1 RFC 1034/1035, DNS
W3C	Web	HTML5 standard CSS standard

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Who's Who in 802.11 (1)

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs (WiFi)
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)

The 802 working groups. The important ones are marked with *.

The ones marked with ↓ are hibernating. The one marked with † gave up and disbanded itself.

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Who's Who in 802.11 (2)

802.13	Unlucky number; nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth, Zigbee)
802.16 *	Broadband wireless (WiMAX)
802.17	Resilient packet ring
802.18	Technical advisory group on radio regulatory issues
802.19	Technical advisory group on coexistence of all these standards
802.20	Mobile broadband wireless (similar to 802.16e)
802.21	Media independent handoff (for roaming over technologies)
802.22	Wireless regional area network

The 802 working groups. The important ones are marked with *.

The ones marked with ↓ are hibernating. The one marked with † gave up and disbanded itself.

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Physical Layer

Guided Transmission media: twisted pairs, coaxial cable, fiber optics, Wireless Transmission.

A closer look at Internet structure

Network edge:

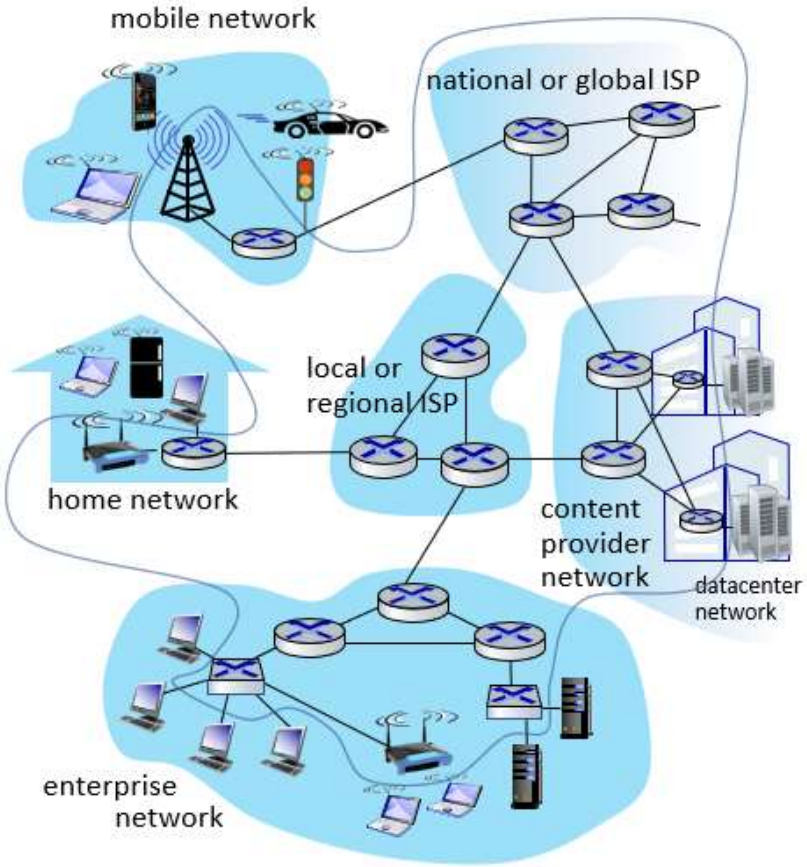
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

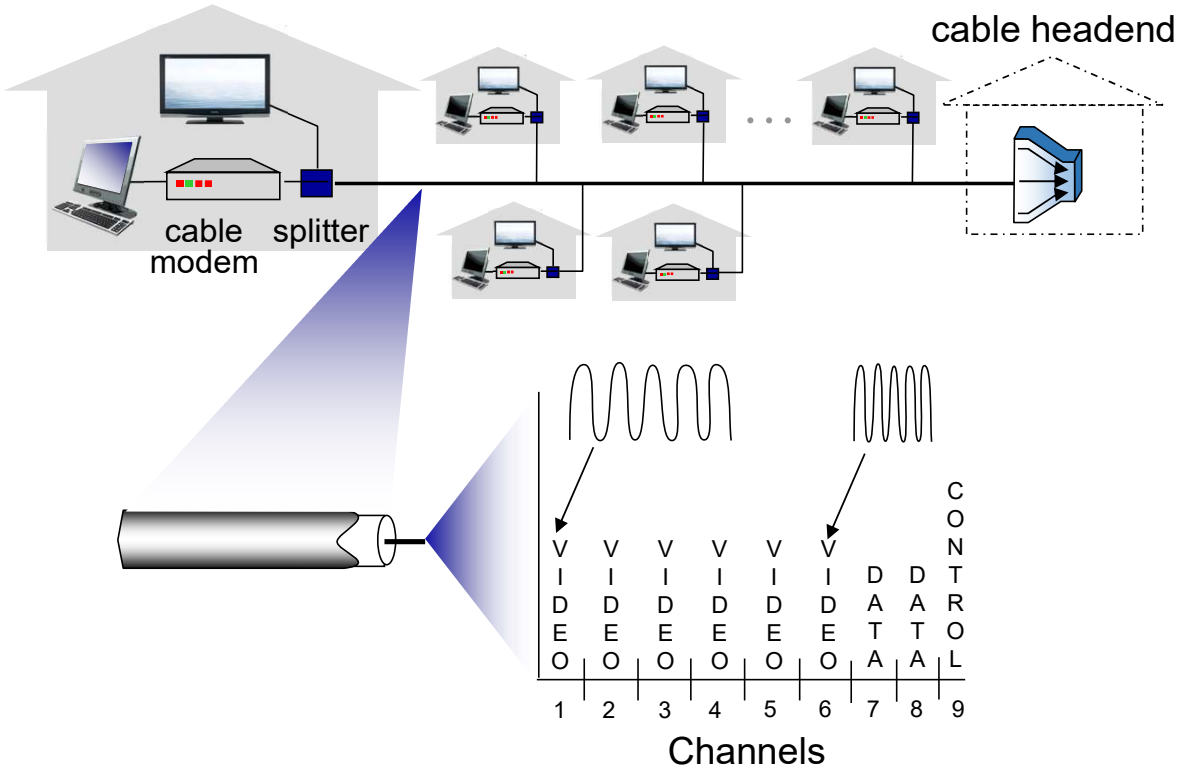
- wired, wireless communication links

Network core:

- interconnected routers
- network of networks

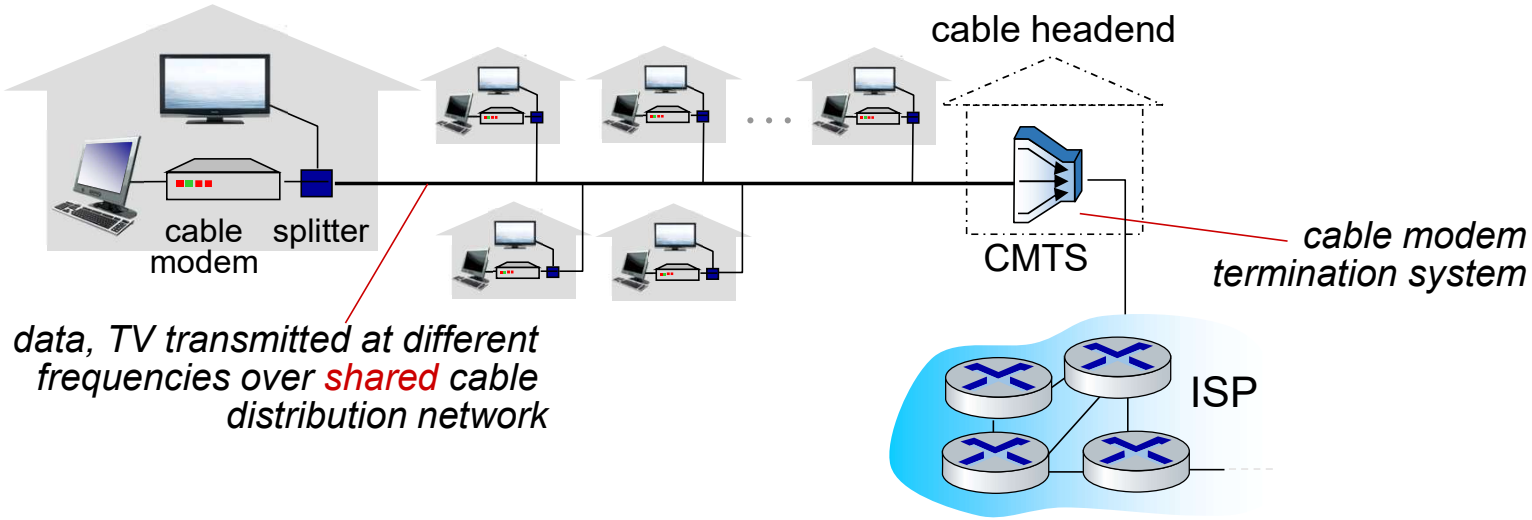


Access networks: cable-based access (Guided Transmission Media)



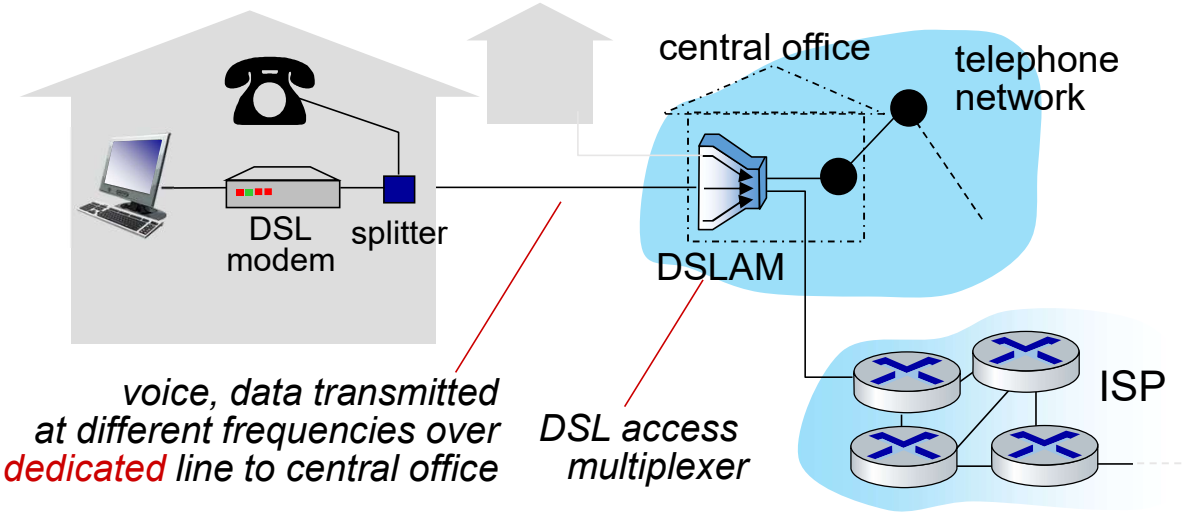
frequency division multiplexing (FDM): different channels transmitted in different frequency bands

Access networks: cable-based access



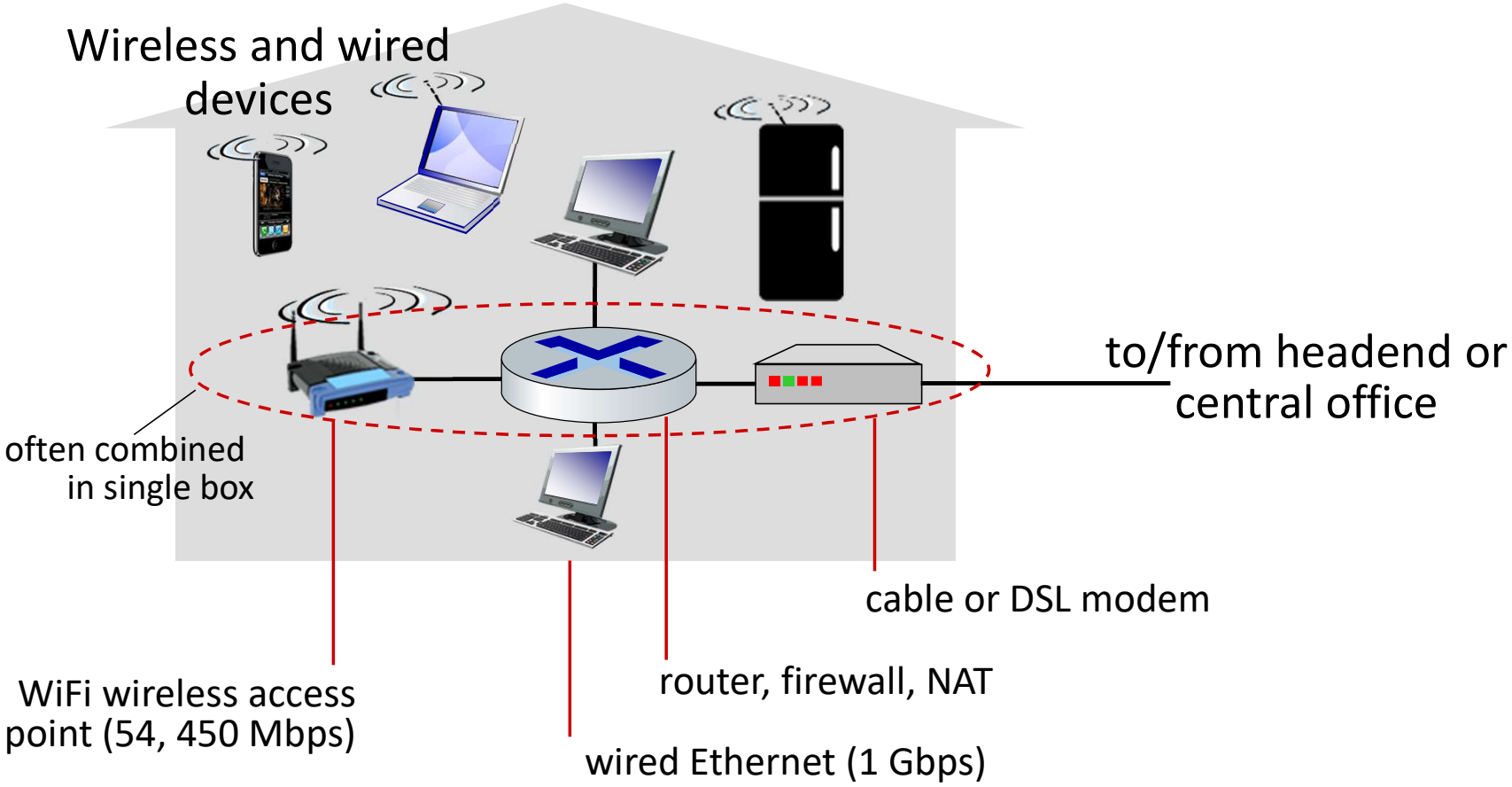
- HFC: hybrid fiber coax
  - asymmetric: up to 40 Mbps – 1.2 Gbps downstream transmission rate, 30-100 Mbps upstream transmission rate
- network of cable, fiber attaches homes to ISP router
  - homes *share access network* to cable headend

Access networks: digital subscriber line (DSL)



- use *existing* telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- 24-52 Mbps dedicated downstream transmission rate
- 3.5-16 Mbps dedicated upstream transmission rate

Access networks: home networks



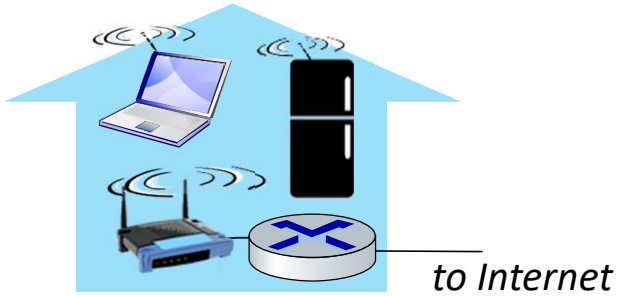
Wireless access networks

Shared *wireless* access network connects end system to router

- via base station aka “access point”

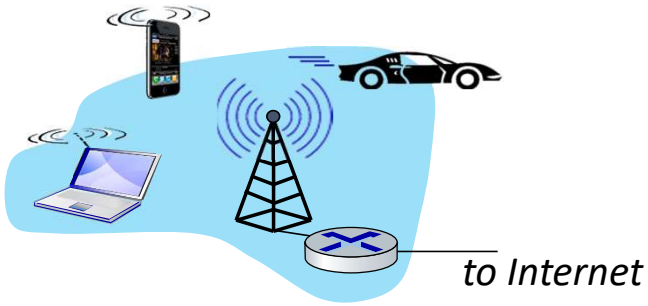
Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate

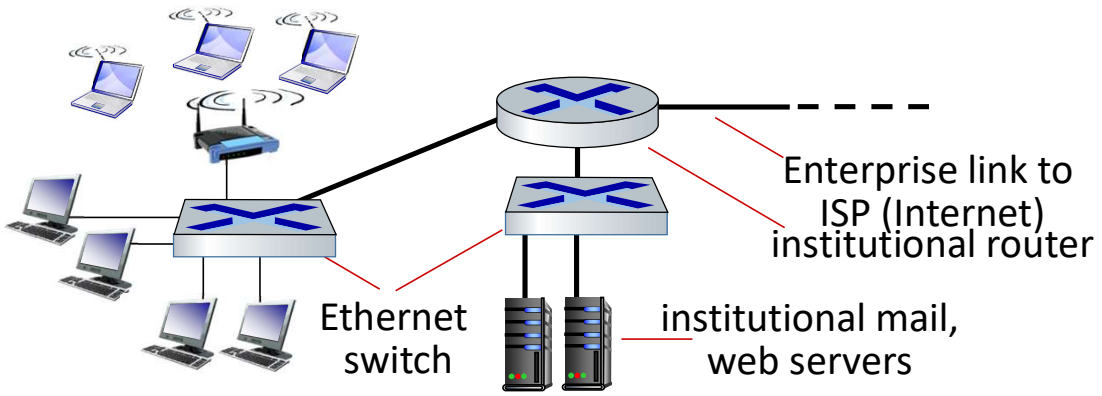


Wide-area cellular access networks

- provided by mobile, cellular network operator (10’s km)
- 10’s Mbps
- 4G/5G cellular networks



Access networks: enterprise networks



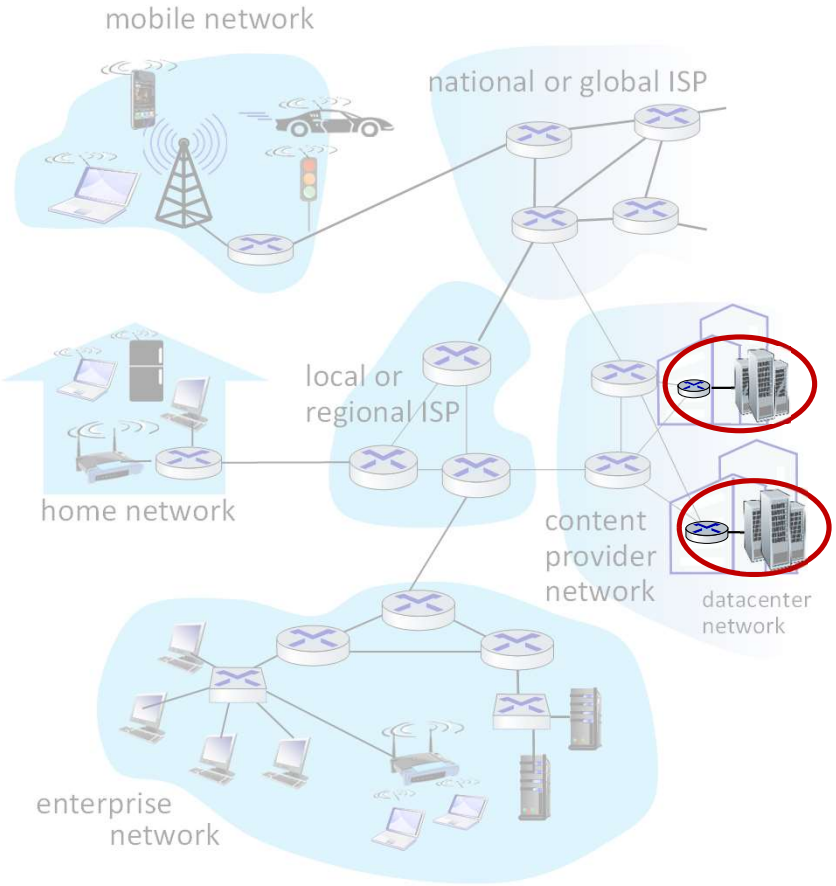
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
  - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
  - WiFi: wireless access points at 11, 54, 450 Mbps

Access networks: data center networks

- high-bandwidth links (10s to 100s Gbps) connect hundreds to thousands of servers together, and to Internet



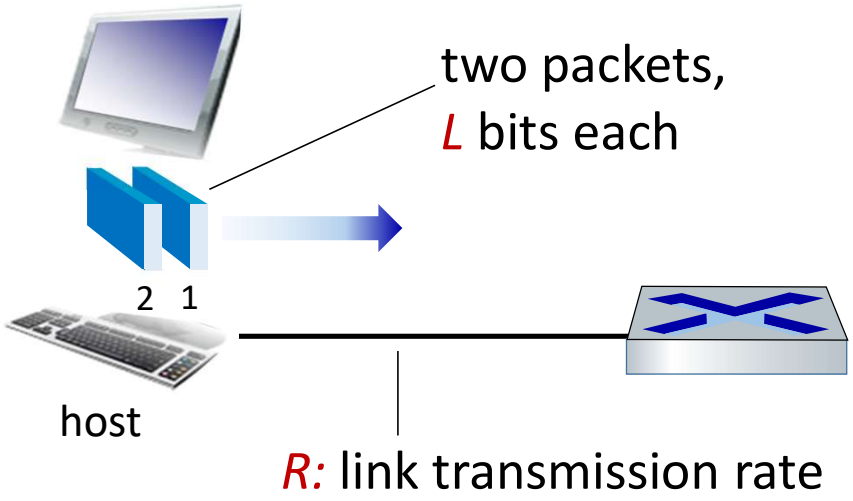
Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



Host: sends *packets* of data

host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length *L* bits
- transmits packet into access network at *transmission rate R*
  - link transmission rate, aka link *capacity, aka link bandwidth*



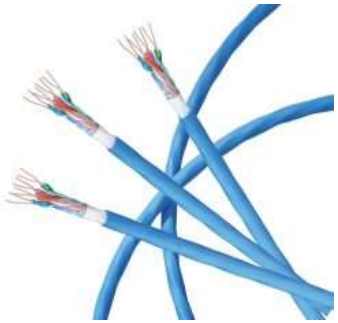
$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Links: physical media

- bit: propagates between transmitter/receiver pairs
- physical link: what lies between transmitter & receiver
- guided media:
  - signals propagate in solid media: copper, fiber, coax
- unguided media:
  - signals propagate freely, e.g., radio

Twisted pair (TP)

- two insulated copper wires
  - Category 5: 100 Mbps, 1 Gbps Ethernet
  - Category 6: 10Gbps Ethernet





Links: physical media

Coaxial cable:

- two concentric copper conductors
- bidirectional
- broadband:
  - multiple frequency channels on cable
  - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise



Links: physical media

Wireless radio

- signal carried in various "bands" in electromagnetic spectrum
- no physical "wire"
- broadcast, "half-duplex" (sender to receiver)
- propagation environment effects:
  - reflection
  - obstruction by objects
  - Interference/noise

Radio link types:

- Wireless LAN (WiFi)
  - 10-100's Mbps; 10's of meters
- wide-area (e.g., 4G/5G cellular)
  - 100's Mbps (4G/5G) over ~10 Km
- Bluetooth: cable replacement
  - short distances, limited rates
- terrestrial microwave
  - point-to-point; 45 Mbps channels
- satellite
  - up to < 100 Mbps (Starlink) downlink
  - 270 msec end-end delay (geostationary)

Data Link Layer (DLL) – L2

Design issues, framing, Error detection and correction.

Data Link Layer Design Issues

- Network layer services
- Framing
- Error control
- Flow control

Data Link Layer

- Algorithms for achieving:
  - Reliable, +
  - Efficient,
    communication of a whole units – frames (as opposed to bits – Physical Layer) between two machines.
- Two machines are connected by a communication channel that acts conceptually like a wire (e.g., telephone line, coaxial cable, or wireless channel).
- Essential property of a channel that makes it “wire-like” connection is that the bits are delivered in exactly the same order in which they are sent.

Data Link Layer

- For ideal channel (no distortion, unlimited bandwidth and no delay) the job of data link layer would be trivial.

- However, limited bandwidth, distortions and delay makes this job very difficult.

Data Link Layer Design Issues

- Physical layer delivers bits of information to and from data link layer. The functions of Data Link Layer are:
  1. Providing a well-defined service interface to the network layer.
  2. Dealing with transmission errors.
  3. Regulating the flow of data so that slow receivers are not swamped by fast senders.

Data Link layer

- Takes the packets from Network Layer, and
- Encapsulates them into **frames**

Data Link Layer Design Issues

- Each frame has a
  - frame header – a field for holding the packet, and
  - frame trailer.
- Frame Management is what Data Link Layer does.

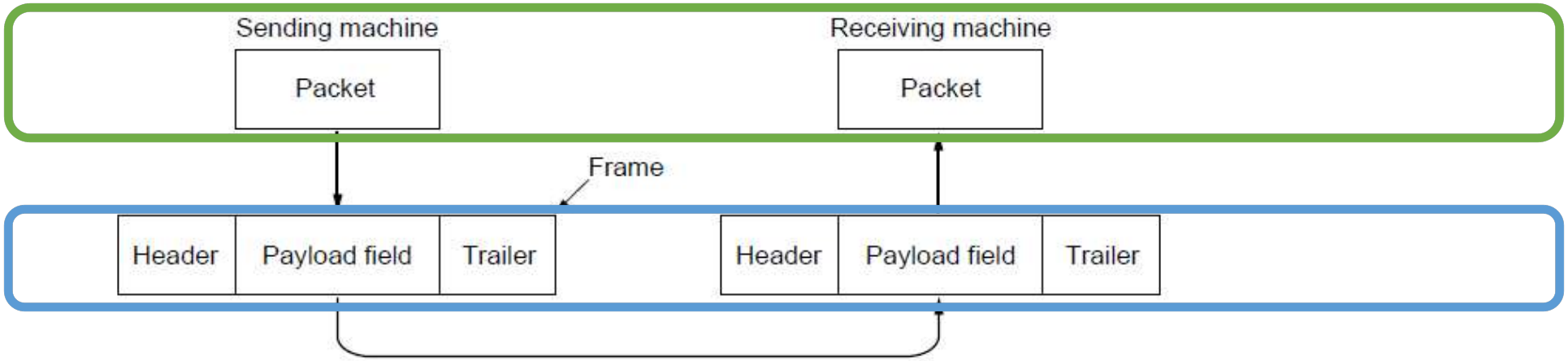


Figure: Relationship between packets and frames.

Services Provided to the Network Layer

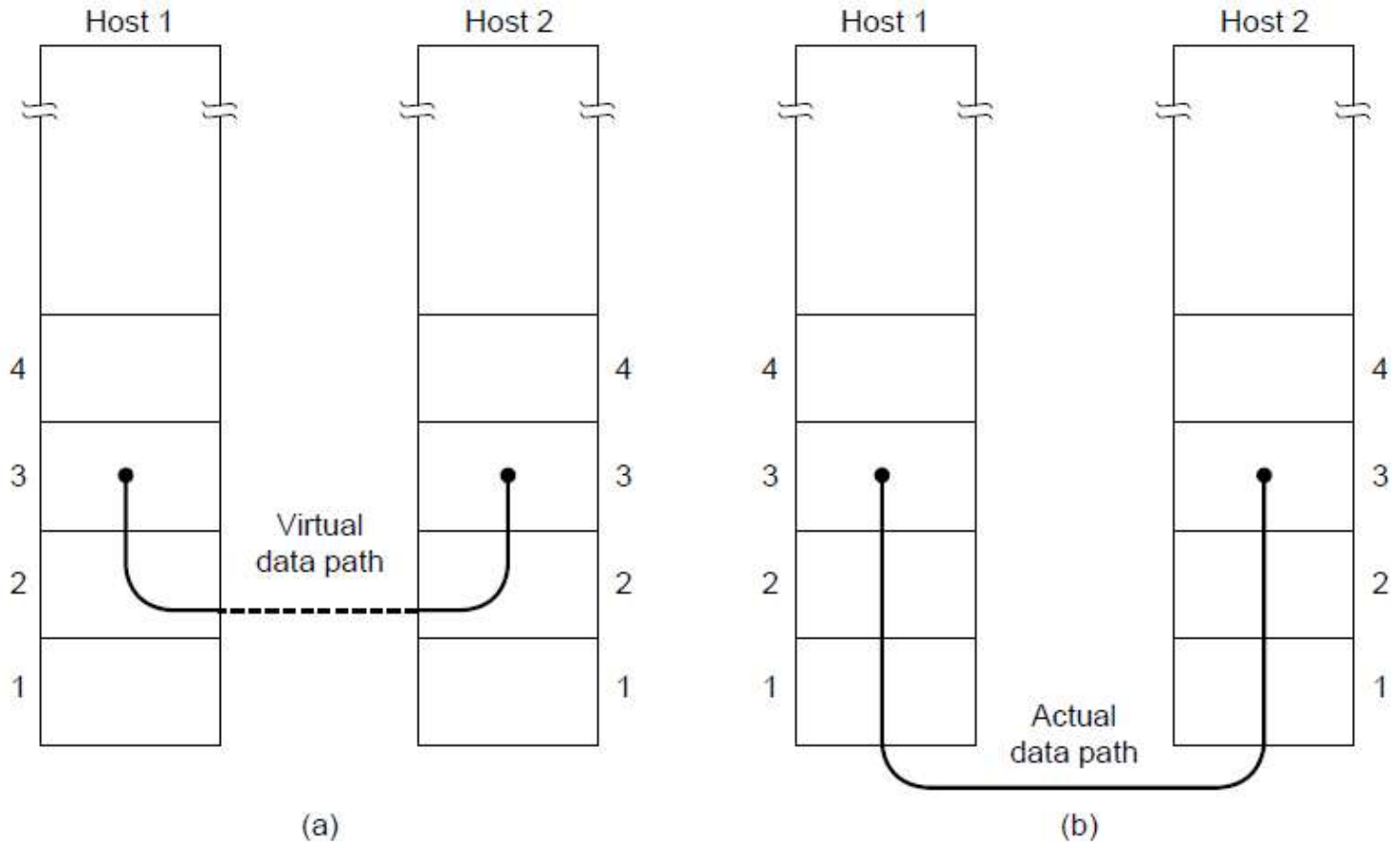
- Principal Service Function of the data link layer is to transfer the data from the network layer on the source machine to the network layer on the destination machine.

  - Process in the network layer that hands some bits to the data link layer for transmission.

  - Job of data link layer is to transmit the bits to the destination machine so they can be handed over to the network layer there (see figure in the next slide).

Network Layer Services

(a) Virtual communication. (b) Actual communication.



Possible Services Offered

1. Unacknowledged connectionless service.

2. Acknowledged connectionless service.

3. Acknowledged connection-oriented service.

Unacknowledged Connectionless Service

- It consists of having the source machine send independent frames to the destination machine without having the destination machine acknowledge them.

- Example: **Ethernet**, **Voice over IP**, etc. in all the communication channel were real time operation is more important that quality of transmission.

Acknowledged Connectionless Service

- Each frame send by the Data Link layer is acknowledged and the sender knows if a specific frame has been received or lost.
- Typically the protocol uses a specific time period that if has passed without getting acknowledgment it will re-send the frame.
- This service is useful for commutation when an unreliable channel is being utilized (e.g., **802.11 WiFi**).

- Network layer does not know frame size of the packets and other restriction of the data link layer. Hence it becomes necessary for data link layer to have some mechanism to optimize the transmission.

Acknowledged Connection Oriented Service

- Source and Destination establish a connection first.
- Each frame sent is numbered
  - Data link layer guarantees that each frame sent is indeed received.
  - It guarantees that each frame is received only once and that all frames are received in the correct order.
- Examples:
  - Satellite channel communication,
  - Long-distance telephone communication, etc.

Acknowledged Connection Oriented Service

- Three distinct phases:
  1. Connection is established by having both side initialize variables and counters needed to keep track of which frames have been received and which ones have not.
  2. One or more frames are transmitted.
  3. Finally, the connection is released – freeing up the variables, buffers, and other resources used to maintain the connection.

Framing

- To provide service to the network layer the data link layer must use the service provided to it by physical layer.
- Stream of data bits provided to data link layer is not guaranteed to be without errors.
- Errors could be:
  - Number of received bits does not match number of transmitted bits (deletion or insertion)
  - Bit Value
- It is up to data link layer to correct the errors if necessary.

Framing

- Transmission of the data link layer starts with breaking up the bit stream
  - into discrete frames
  - Computation of a checksum for each frame, and
  - Include the checksum into the frame before it is transmitted.
- Receiver computes its checksum error for a receiving frame and if it is different from the checksum that is being transmitted will have to deal with the error.

- Framing is more difficult than one could think!

Framing Methods

1. Byte count.
2. Flag bytes with byte stuffing.
3. Flag bits with bit stuffing.
4. Physical layer coding violations.

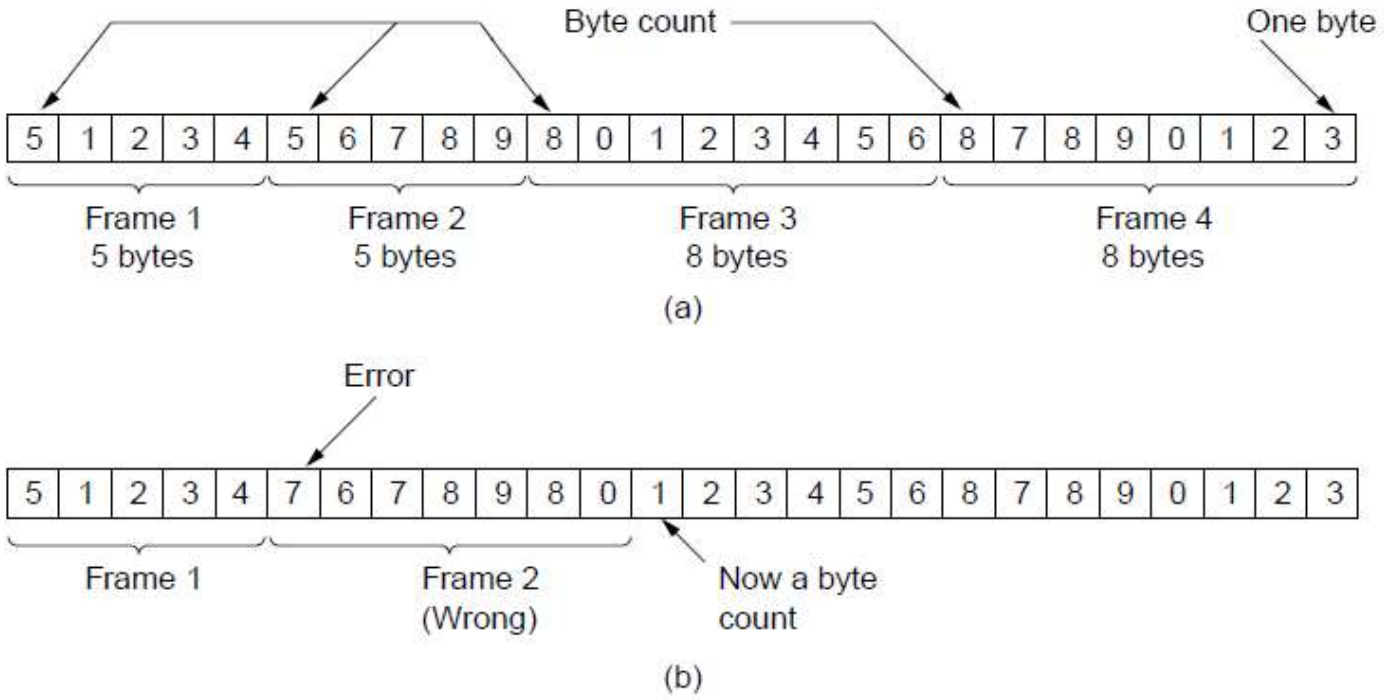
Byte Count Framing Method

- It uses a field in the header to specify the number of bytes in the frame.
- Once the header information is being received it will be used to determine end of the frame.
- See figure in the next slide:
- Trouble with this algorithm is that when the count is incorrectly received the destination will get out of synch with transmission.

    - Destination may be able to detect that the frame is in error but it does not have a means (in this algorithm) how to correct it.

Framing (1)

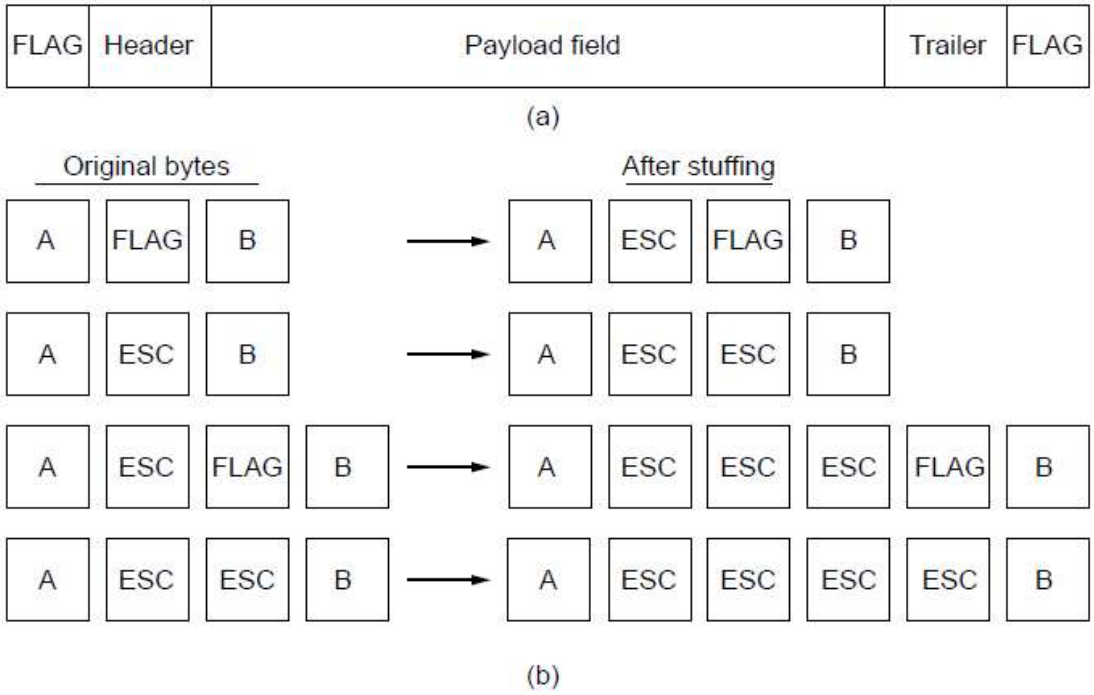
A byte stream. (a) Without errors. (b) With one error.



Flag Bytes with Byte Stuffing Framing Method

- This methods gets around the boundary detection of the frame by having each appended by the frame start and frame end special bytes.
- If they are the same (beginning and ending byte in the frame) they are called **flag byte**.
- In the next slide figure this byte is shown as FLAG.
- If the actual data contains a byte that is identical to the FLAG byte (e.g., picture, data stream, etc.) the convention that can be used is to have escape character inserted just before the “FLAG” character.

Framing (2)

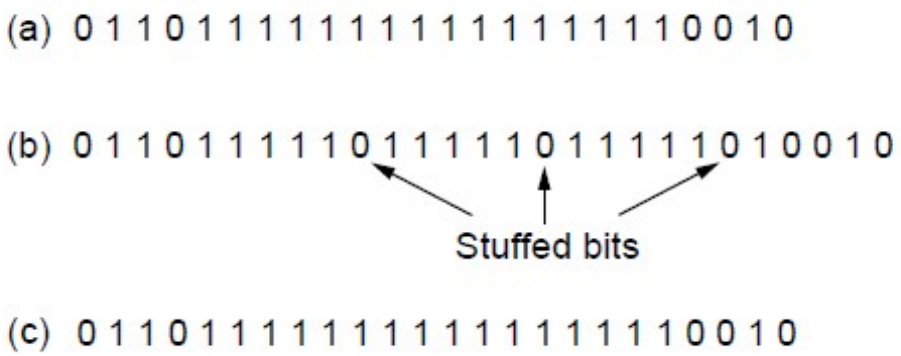


- A frame delimited by flag bytes.
- Four examples of byte sequences before and after byte stuffing.

Flag Bits with Bit Stuffing Framing Method

- This methods achieves the same thing as Byte Stuffing method by using Bits (1) instead of Bytes (8 Bits).
- It was developed for High-level Data Link Control (HDLC) protocol.
- Each frames begins and ends with a special bit pattern:
  - 01111110 or 0x7E <- Flag Byte
  - Whenever the sender's data link layer encounters five consecutive 1s in the data it automatically stuffs a 0 bit into the outgoing bit stream.
  - USB uses bit stuffing.

Framing (3)



Bit stuffing. (a) The original data. (b) The data as they appear on the line. (c) The data as they are stored in the receiver's memory after destuffing.

Framing

- Many data link protocols use a combination of presented methods for safety. For example in Ethernet and 802.11 each frame begin with a well-defined pattern called a preamble.
- Preamble is typically 72 bits long.
- It is then followed by a length fileld.

Error Control

- After solving the marking of the frame with start and end the data link layer has to handle eventual errors in transmission or detection.
  - Ensuring that all frames are delivered to the network layer at the destination and in proper order.
- Unacknowledged connectionless service: it is OK for the sender to output frames regardless of its reception.
- Reliable connection-oriented service: it is NOT OK.

Error Control

- Reliable connection-oriented service usually will provide a sender with some feedback about what is happening at the other end of the line.
  - Receiver Sends Back Special Control Frames.
  - If the Sender Receives positive Acknowledgment it will know that the frame has arrived safely.
- Timer and Frame Sequence Number for the Sender is Necessary to handle the case when there is no response (positive or negative) from the Receiver .

Flow Control

- Important Design issue for the cases when the sender is running on a fast powerful computer and receiver is running on a slow low-end machine.
- Two approaches:

1. **Feedback-based flow control**

2. **Rate-based flow control**

Feedback-based Flow Control

- Receiver sends back information to the sender giving it permission to send more data, or
- Telling sender how receiver is doing.

Rate-based Flow Control

- Built in mechanism that limits the rate at which sender may transmit data, without the need for feedback from the receiver.

Error Detection and Correction

- Two basic strategies to deal with errors:
  1. Include enough redundant information to enable the receiver to deduce what the transmitted data must have been.

     Error correcting codes or FEC.
  2. Include only enough redundancy to allow the receiver to deduce that an error has occurred (but not which error).

     Error detecting codes.

Error Detection and Correction

- Error codes are examined in Link Layer because this is the first place that we have run up against the problem of reliability transmitting groups of bits.
  - Codes are reused because reliability is an overall concern.
  - The error correcting code are also seen in the physical layer for noise channels.
  - Commonly they are used in link, network and transport layer.

Error Detection and Correction

- Error codes have been developed after long fundamental research conducted in mathematics.
- Many protocol standards get codes from the large field in mathematics.

Error Detection & Correction Code (1)

1. Hamming codes.
2. Binary convolutional codes.
3. Reed-Solomon codes.
4. Low-Density Parity Check codes.

Error Detection & Correction Code

- All the codes presented in previous slide add redundancy to the information that is being sent.
- A frame consists of
  - m data bits (message) and
  - r redundant bits (check).
- ***Block code*** - the r check bits are computed solely as function of the m data bits with which they are associated.

- ***Systemic code*** – the m data bits are send directly along with the check bits.
- ***Linear code*** – the r check bits are computed as a linear function of the m data bits.

Error Detection & Correction Code

- n – total length of a block (i.e., $n = m + r$)
- (n, m) – code
- n – bit ***codeword*** containing n bits.
- m/n – code rate (range ½ for noisy channel and close to 1 for high-quality channel).

Requirement for error detection: A code is an error-detecting code if and only if the minimum distance between any two code words is two.

Requirement for error detection: For a code to be error-correcting, the minimum distance between any two code words must be more than two.

Number of additional bits should be such that it can point the position of bits in the error. If **k** is the number of additional bits then, the condition is $2^k > m+k+1$.

Error Detection & Correction Code

Example

- Transmitted: 10001001
- Received: 10110001

XOR operation gives number of bits that are different.

- XOR: 00111000

- Number of bit positions in which two codewords differ is called ***Hamming Distance***. It shows that two codes are ***d*** distance apart, and it will require ***d*** errors to convert one into the other.

Error Detection & Correction Code

- n – total length of a block (i.e., $n = m + r$)
- (n, m) – code
- n – bit ***codeword*** containing n bits.
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Requirement for error correction: For a code to be error-correcting, the minimum distance between any two code words must be more than two.

Number of additional bits should be such that it can point the position of bits in the error. If **r** is the number of additional bits then, the condition is **$2^r > m+r+1$.**

Error Detection & Correction Code

- All 2^m possible data messages are legal, but due to the way the check bits are computers not all 2^n possible code words are used.
- Only small fraction of $2^m/2^n = 1/2^r$ ***are possible will be legal codewords.***
- The error-detecting and error-correcting codes of the block code depend on this Hamming distance.
- To reliably detect d error, one would need a distance $d+1$ code.
- To correct d error, one would need a distance $2d+1$ code.

Error Detection & Correction Code

No. of Data Bits	No. of Redundancy Bits	Total Number of Bits
1	2	3
2	3	5
3	3	6
4	3	7
5	4	9
6	4	10
7	4	11

Error Detection & Correction Code

Example:

- 4 valid codes:
  - 0000000000
  - 0000011111
  - 1111100000
  - 1111111111
- Minimal Distance of this code is 5 => can correct and double errors and it detect quadruple errors.
- 0000000111 => single or double – bit error. Hence the receiving end must assume the original transmission was 0000011111.
- 0000000000 => had triple error that was received as 0000000111 it would be detected in error.

Error Detection & Correction Code

- One cannot perform double errors and at the same time detect quadruple errors.
- Error correction requires evaluation of each candidate codeword which may be time consuming search.
- Through design this search time can be minimized.
- In theory if n = m + r, this requirement becomes:
  - $(m + r + 1) \leq 2^r$



Note

The Hamming distance between two words is the number of differences between corresponding bits.

The Hamming Code

- Basic Approach:

1. To each group of m information bits, r parity bits are added to form (m+r) bit code.
2. Location of each of the (m+r) digits is assigned a decimal value.
3. The r parity bits are placed in positions, 1, 2, ….., 2^{r-1} positions.
4. R parity checks are performed on selected digits of each codeword.
5. At the receiving end, the parity bits are recalculated. The decimal value of the r parity bits provides the bit-position in error, if any.

The Hamming Code

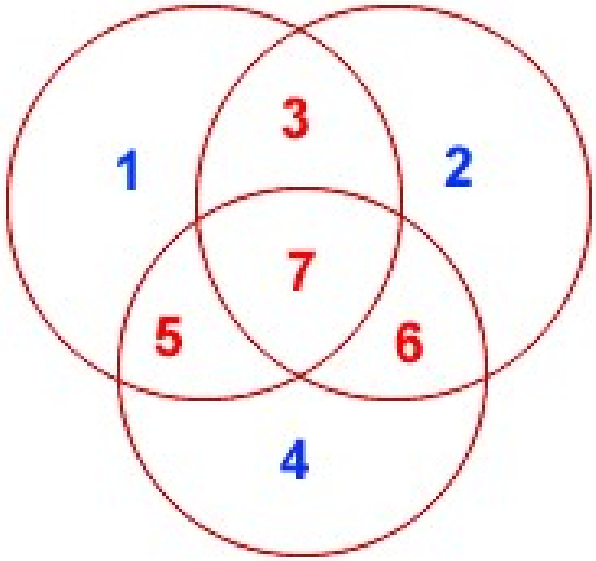
Consider a message having four data bits (D) which is to be transmitted as a 7-bit codeword by adding three error control bits. This would be called a (7,4) code. The three bits to be added are three EVEN Parity bits (P), where the parity of each is computed on different subsets of the message bits as shown below.

7	6	5	4	3	2	1	
D	D	D	P	D	P	P	7-BIT CODEWORD
D	-	D	-	D	-	P	(EVEN PARITY)
D	D	-	-	D	P	-	(EVEN PARITY)
D	D	D	P	-	-	-	(EVEN PARITY)

Hamming Code

- **Why Those Bits?** - The three parity bits (1,2,4) are related to the data bits (3,5,6,7) as shown at right. In this diagram, each overlapping circle corresponds to one parity bit and defines the four bits contributing to that parity computation. For example, data bit 3 contributes to parity bits 1 and 2. Each circle (parity bit) encompasses a total of four bits, and each circle must have EVEN parity. Given four data bits, the three parity bits can easily be chosen to ensure this condition.
- It can be observed that changing any one bit numbered 1..7 uniquely affects the three parity bits. Changing bit 7 affects all three parity bits, while an error in bit 6 affects only parity bits 2 and 4, and an error in a parity bit affects only that bit. The location of any single bit error is determined directly upon checking the three parity circles.

Hamming Code



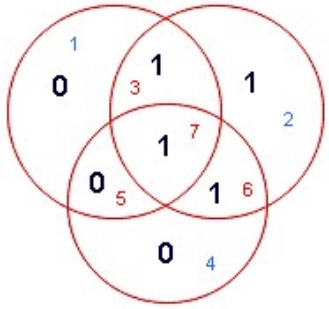
Hamming Code

- For example,
- the message 1010

Error Positions	Position Numbers
0(no error)	0 0 0
1	0 0 1
2	0 1 0
3	0 1 1
4	1 0 0
5	1 0 1
6	1 1 0
7	1 1 1

Hamming Codes

- When these seven bits are entered into the parity circles, it can be confirmed that the choice of these three parity bits ensures that the parity within each circle is EVEN, as shown here.



Hamming Code

- It may now be observed that if an error occurs in any of the seven bits, that error will affect different combinations of the three parity bits depending on the bit position.

- For example, suppose the above message 1100110 is sent and a single bit error occurs such that the codeword 1110110 is received:

transmitted message		received message
1 1 0 0 1 1 0	---------->	1 1 1 0 1 1 0
BIT: 7 6 5 4 3 2 1		BIT: 7 6 5 4 3 2 1

The above error (in bit 5) can be corrected by examining which of the three parity bits was affected by the bad bit:

Hamming Code

7	6	5	4	3	2	1			
1	1	1	**0**	1	**1**	**0**	7-BIT CODEWORD		
1	-	1	-	1	-	0	(EVEN PARITY)	**NOT!**	**1**
1	1	-	-	1	1	-	(EVEN PARITY)	**OK!**	**0**
1	1	1	0	-	-	-	(EVEN PARITY)	**NOT!**	**1**

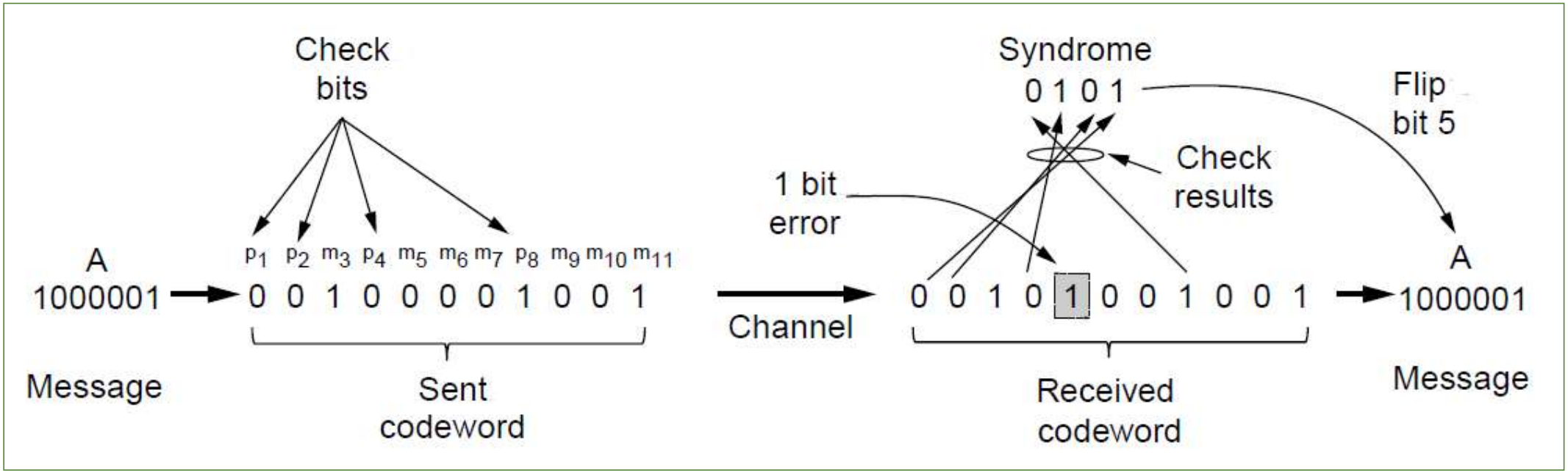
Hamming Code

- *In fact, the bad parity bits labeled **101** point directly to the bad bit since **101** binary equals **5**.* Examination of the 'parity circles' confirms that any single bit error could be corrected in this way.
- The value of the Hamming code can be summarized:

1. Detection of 2 bit errors (assuming no correction is attempted);
2. Correction of single bit errors;
3. Cost of 3 bits added to a 4-bit message.

- The ability to correct single bit errors comes at a cost which is less than sending the entire message twice. (Recall that simply sending a message twice accomplishes no error correction.)

Error Detection Codes (2)

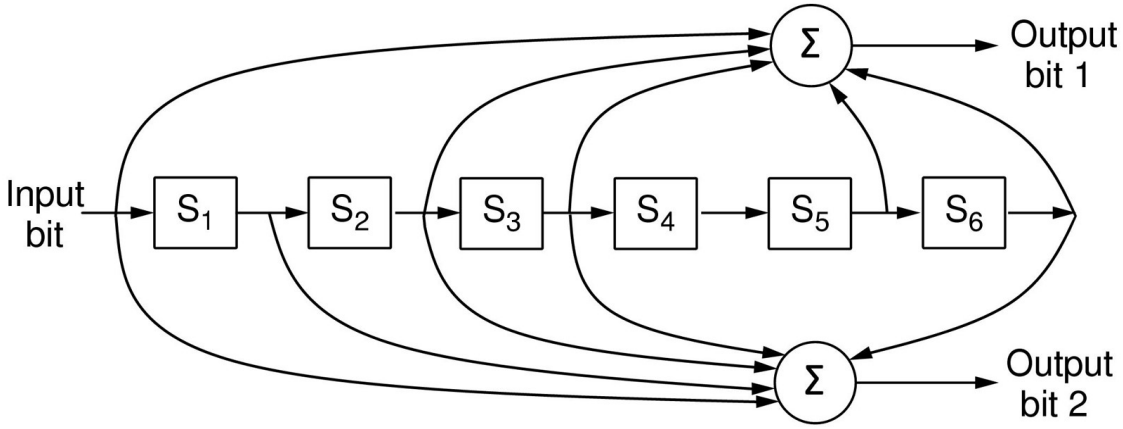


Example of an (11, 7) Hamming code correcting a single-bit error.

Convolutional Codes

- Not a block code
- There is no natural message size or encoding boundary as in a block code.
- The output depends on the current and previous input bits. Encoder has memory.
- The number of previous bits on which the output depends is called the **constraint length** of the code.
- They are deployed as part of the
  - GSM mobile phone system
  - Satellite Communications, and
  - 802.11

Error Detection Codes (3)



The NASA binary convolutional code used in 802.11.

Convolutional Encoders

- Like any error-correcting code, a convolutional code works by adding some structured redundant information to the user's data and then correcting errors using this information.
- A convolutional encoder is a *linear system*.
- A binary convolutional encoder can be represented as a *shift register*. The outputs of the encoder are modulo 2 sums of the values in the certain register's cells. The input to the encoder is either the unencoded sequence (for *non-recursive codes*) or the unencoded sequence added with the values of some register's cells (for *recursive codes*).
- Convolutional codes can be *systematic* and *non-systematic*. Systematic codes are those where an unencoded sequence is a part of the output sequence. Systematic codes are almost always recursive, conversely, non-recursive codes are almost always non-systematic.

Convolutional Encoders

- A combination of register's cells that forms one of the output streams (or that is added with the input stream for recursive codes) is defined by a *polynomial*. Let *m* be the maximum degree of the polynomials constituting a code, then *K*=*m*+1 is a *constraint length* of the code.

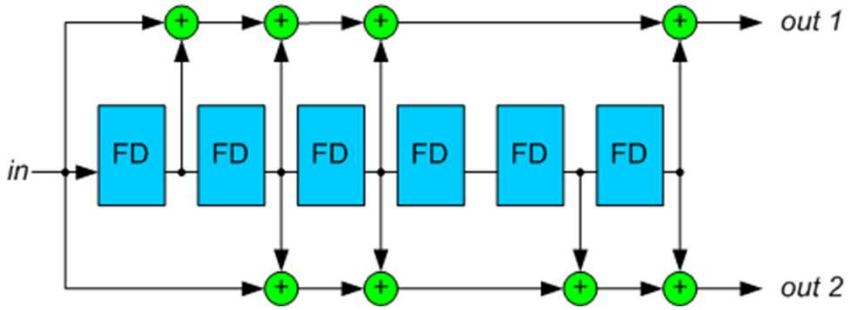


Figure 1. A standard NASA convolutional encoder with polynomials (171,133).

Convolutional Encoders

- For example, for the decoder on the Figure 1, the polynomials are:

$$g_1(z)=1+z+z^2+z^3+z^6$$

$$g_2(z)=1+z^2+z^3+z^5+z^6$$

- A code rate is an inverse number of output polynomials.
- For the sake of clarity, in this article we will restrict ourselves to the codes with rate R=1/2. Decoding procedure for other codes is similar.
- Encoder polynomials are usually denoted in the octal notation. For the above example, these designations are “1111001” = 171 and “1011011” = 133.
- The constraint length of this code is 7.
- An example of a recursive convolutional encoder is on the Figure 2.

Example of the Convolutional Encoder

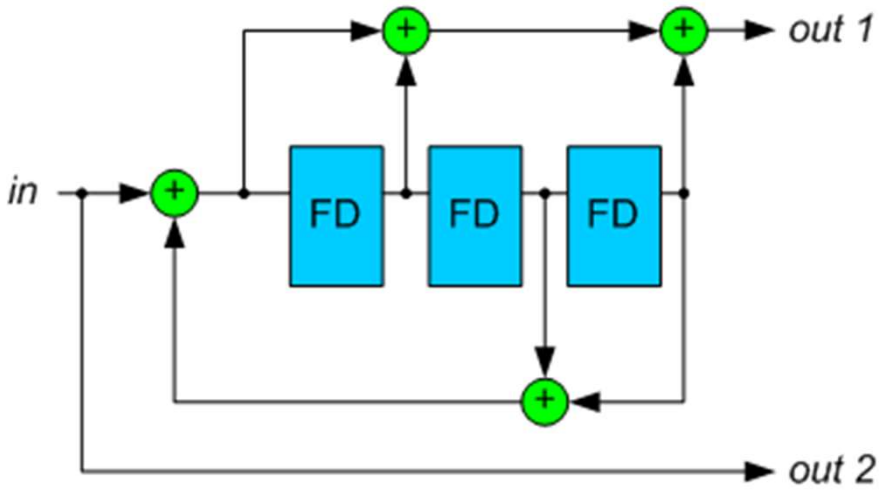


Figure 2. A recursive convolutional encoder.

Trellis Diagram

- A convolutional encoder is often seen as a *finite state machine*. Each state corresponds to some value of the encoder's register. Given the input bit value, from a certain state the encoder can move to two other states. These state transitions constitute a diagram which is called a *trellis diagram*.

- A trellis diagram for the code on the Figure 2 is depicted on the Figure 3. A solid line corresponds to input 0, a dotted line – to input 1 (note that encoder states are designated in such a way that the rightmost bit is the newest one).

- Each path on the trellis diagram corresponds to a valid sequence from the encoder's output. Conversely, any valid sequence from the encoder's output can be represented as a path on the trellis diagram. One of the possible paths is denoted as red (as an example).

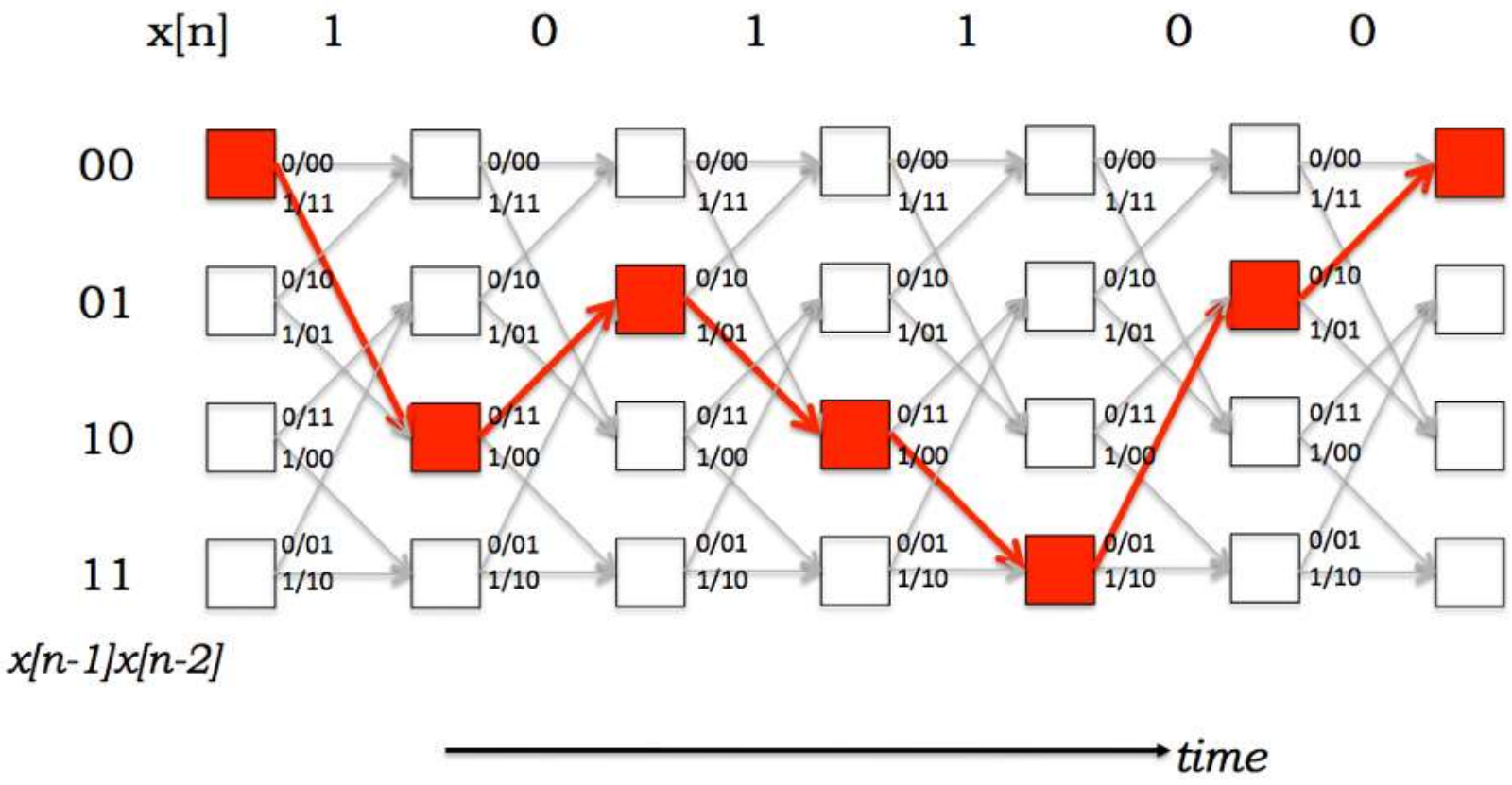


Figure 8-5: The trellis is a convenient way of viewing the decoding task and understanding the time evolution of the state machine.

Trellis Diagram

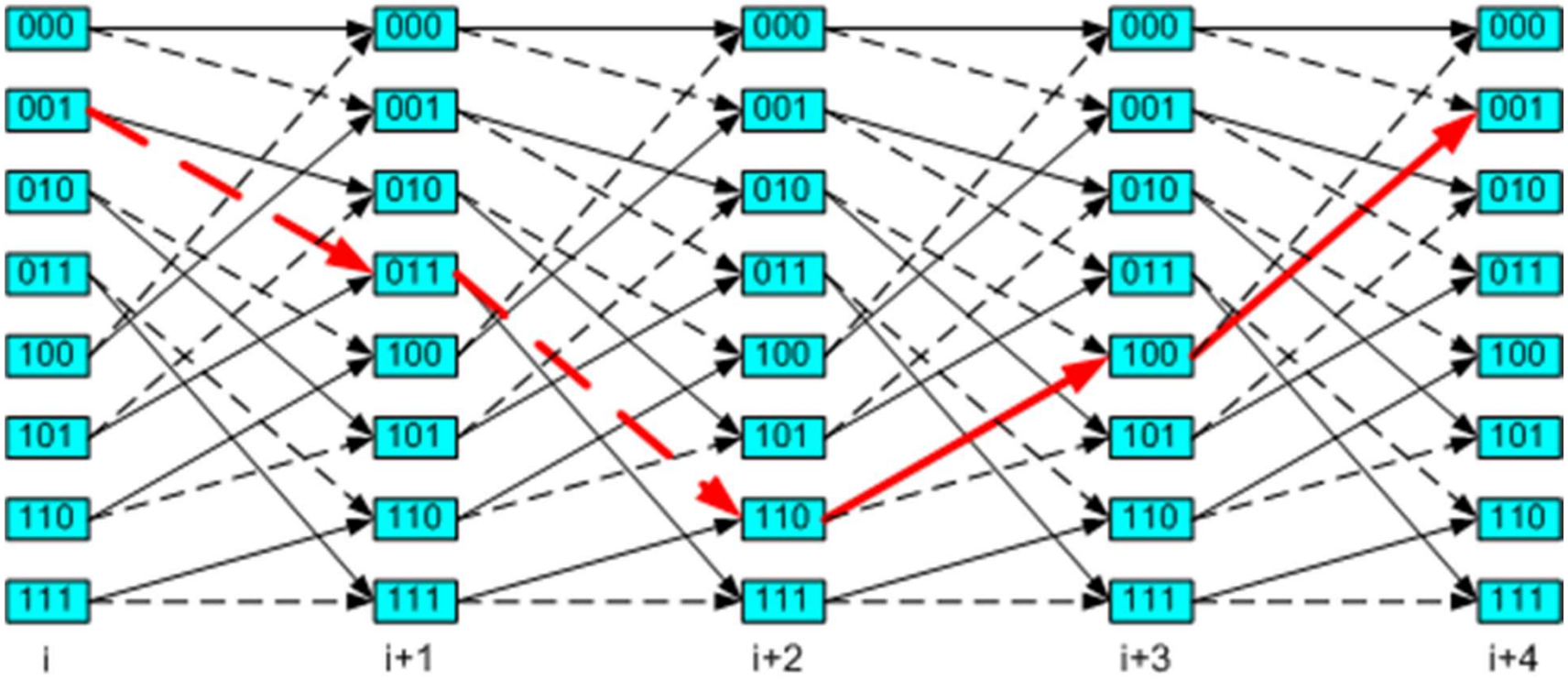


Figure 3. A trellis diagram corresponding to the encoder on the Figure 2.

Trellis Diagram

- Note that each state transition on the diagram corresponds to a pair of output bits.
- There are only two allowed transitions for every state, so there are two allowed pairs of output bits, and the two other pairs are forbidden.
- If an error occurs, it is very likely that the receiver will get a set of forbidden pairs, which don't constitute a path on the trellis diagram.
- So, the task of the decoder is to find a path on the trellis diagram which is the closest match to the received sequence.

Trellis Diagram

- Let's define a *free distance* d_f as a minimal Hamming distance between two different allowed binary sequences (a Hamming distance is defined as a number of differing bits).
- A free distance is an important property of the convolutional code. It influences a number of closely located errors the decoder is able to correct.

Viterbi Algorithm

- Viterbi algorithm reconstructs the maximum-likelihood path for a given input sequence.

Reed-Solomon Code

- Reed-Solomon (RS) codes are **block error-correcting codes**.
- Used to **detect and correct multiple symbol errors** in a block of data.

Key Features

- Operate on **symbols** (usually 1 byte = 8 bits), **not individual bits**.
- Can correct up to t symbol errors in a block if 2t check symbols are added.
- Known as **Maximum Distance Separable (MDS)** codes.

How They Work

- Encode data as a **polynomial over a finite field** (Galois Field, GF).
- Add redundant symbols by evaluating the polynomial at additional points.
- At the receiver: Use decoding algorithms to correct errors (e.g., Berlekamp-Massey algorithm).

Reed-Solomon Code

Why Reed-Solomon?

- Excellent for **burst error correction** (common in physical media and noisy links).
- Particularly useful when:
  - **Retransmission is costly or impossible** (e.g., satellite links).
  - Data is transmitted once only (e.g., broadcast, CD/DVD).

Real-World Applications

- **CDs/DVDs** – to recover data from scratches.
- **QR Codes** – allow scanning even if partially damaged.
- **Digital TV**, **DSL**, **Wi-Fi (802.11)**.
- **Satellite and Space Communication**.

Reed-Solomon Code

Comparison with Other Codes

Code Type	Works On	Error Type Handled	Efficiency
Hamming Code	Bits	Single-bit errors	Low
CRC	Bits	Error detection only	High
Convolution Codes	Bits	Single-bit errors	High
Reed-Solomon	Symbols (Bytes)	Burst/multiple errors	Moderate

Limitations

- Computationally heavier than simpler codes.
- Requires symbol-level operations over Galois Fields.

Low-Density Parity-Check (LDPC)

What Are LDPC Codes?

- **Linear error-correcting codes** that use a sparse parity-check matrix.
- Designed to **detect and correct multiple errors** with high efficiency.
- Known for being **very close to the Shannon limit** (optimal error correction).

Key Features

- **"Low Density"** = few 1s in the parity-check matrix
- Efficient for **long messages**
- Can be **decoded iteratively** using **belief propagation** (message passing)

Structure

- Defined by a **parity-check matrix (H)**:
  - Large, but sparse (mostly 0s)
  - Example: $H \times x^t = 0 \pmod 2$
- Matrix ensures **only a few bits** participate in each parity check.

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  - Example: $H \times x^t = 0 \pmod 2$
- Matrix ensures **only a few bits** participate in each parity check.

Low-Density Parity-Check (LDPC)

- **Why LDPC Codes?**
  - Very **low error rates** in high-noise environments
  - Outperform older codes (like Reed-Solomon) for **longer block lengths**
  - Used in **high-speed communication systems** like:
    - **Wi-Fi (802.11n/ac/ax)**
    - **5G cellular networks**
    - **Satellite and deep-space communication**
- **Decoding Method**
  - **Iterative decoding** using the **belief propagation algorithm**
  - Involves repeated message exchange between:
    - **Variable nodes** (data bits)
    - **Check nodes** (parity checks)

Low-Density Parity-Check (LDPC)

Real-World Applications

- **Wi-Fi standards (802.11n/ac)**
- **Digital Video Broadcasting (DVB-S2)**
- **5G NR** (NextGen wireless)
- **Space communication** (NASA, ESA)

Comparison with Other Codes

Feature	Reed-Solomon	LDPC
Error Type	Burst, symbol	Random, long block
Efficiency	High (short block)	Higher (long block)
Used in	CDs, QR, DVB	Wi-Fi, 5G, Satellite

Low-Density Parity-Check (LDPC)

Limitations

- Works best with **long messages**
- **Complex decoder implementation** compared to simpler FEC codes

Summary

- LDPC codes are **powerful, efficient** error-correcting codes using sparse matrices.
- Extremely useful for **high-throughput, low-latency communication**.
- Approaches theoretical limits of error correction performance.

Error-Detecting Codes (1)

1. Parity.

2. Checksums.

3. Cyclic Redundancy Checks (CRCs).

Parity

- A parity bit is a simple form of error detection used in digital communications, computing, and data storage.
- It is an extra bit added to a binary code to ensure the accuracy of data transmission or storage.
- The value of the parity bit is determined based on the number of ones (or zeros) in the data being transmitted.
- The number has “odd parity” if it contains an odd number of 1-bits and is “even parity” if it contains an even number of 1-bits.

7 bits of data	(count of 1-bits)	8 bits including parity	
		even	odd
0000000	0	00000000	00000001
1010001	3	10100011	10100010
1101001	4	11010010	11010011
1111111	7	11111111	11111110

Parity

- If the block is badly garbled by a long burst error, the probability that the error will be detected is only 0.5, which is hardly acceptable.
- rectangular matrix-send one parity bit for each row.
- Better Protection – **interleaving--**Interleaving is a general technique to convert a code that detects (or corrects) isolated errors into a code that detects (or corrects) burst errors.

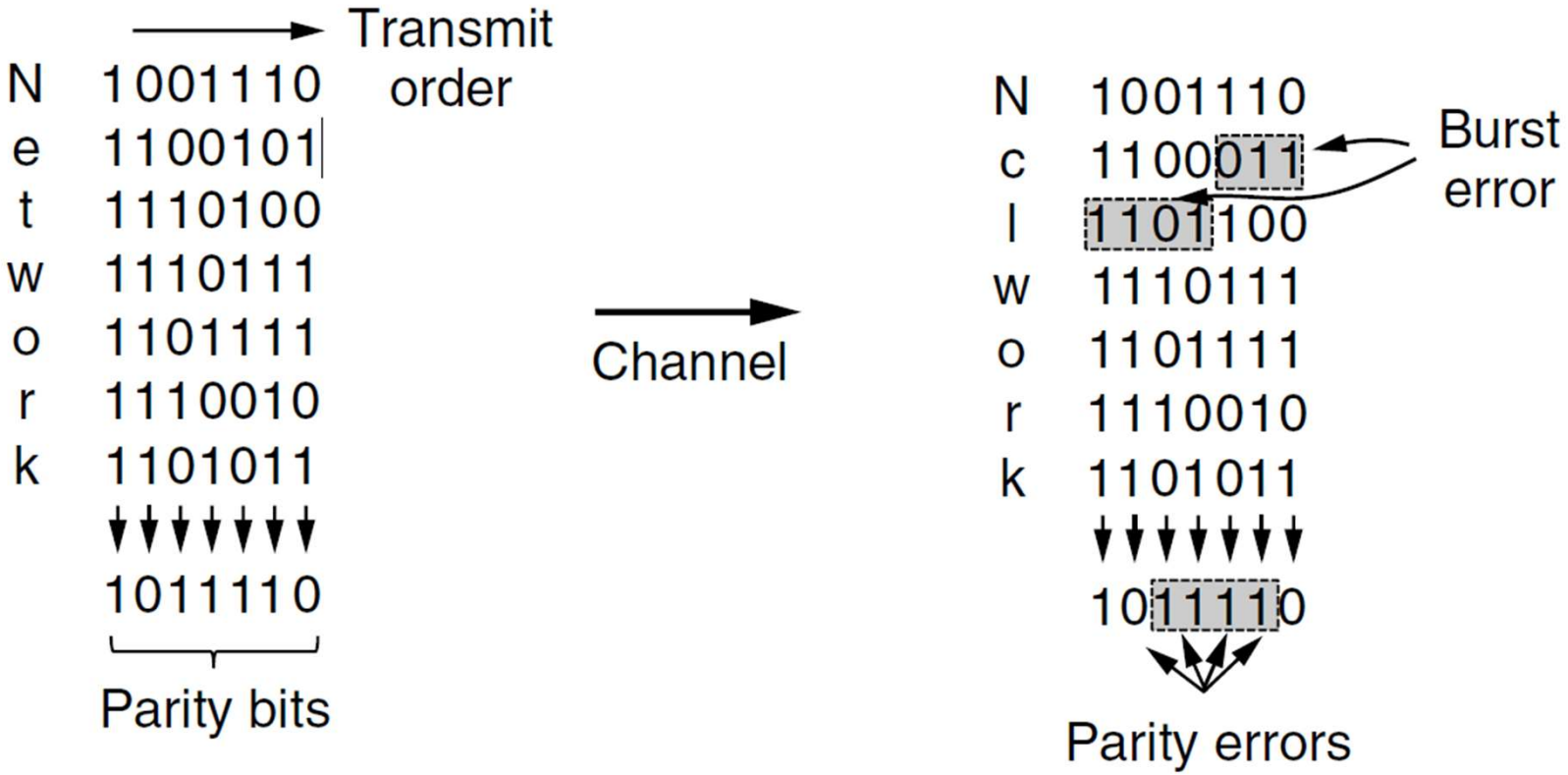
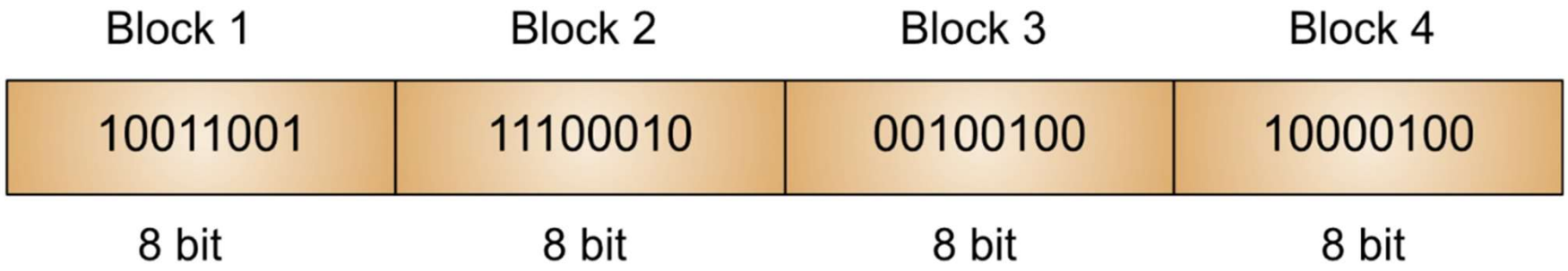


Figure 3-8. Interleaving of parity bits to detect a burst error.

Check Sum

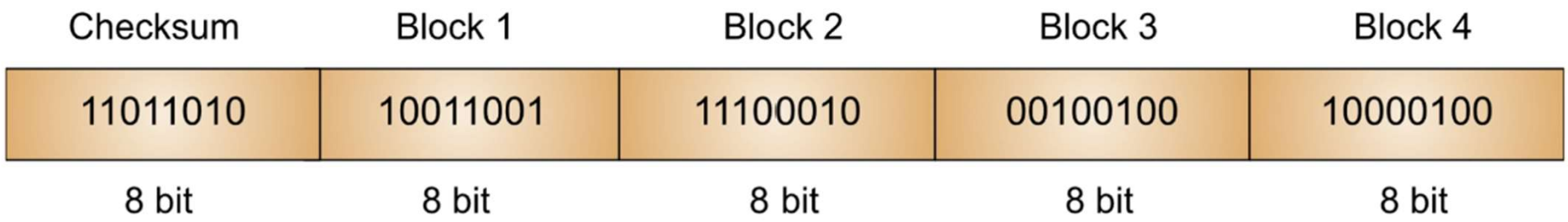
Eg: Data to be sent

10011001 11100010 00100100 10000100.



- Add all the bit using Binary Addition- 1 0 0 0 1 0 0 0 1 1 (answer)
- wrap the carry in the resultant sum- 0 0 1 0 0 1 0 1
- perform the 1's complement- 1 1 0 1 1 0 1 0 (Checksum)

Check Sum



- Sum of all Blocks(including check sum)--1 0 1 1 1 1 1 1 0 1
- Wrap the carry
- do the 1's complement.

Check Sum

- **Sender's Side**
- At the sender's side, the following steps take place:

1. Break the message into 'k' number of blocks and 'n' bit for each block.

2. Add all the 'k' data blocks.

3. Suppose, while adding, you encountered a carry, add it to the sum of the 'k' data blocks.

4. To get the checksum, do the 1's complement.

- **Data Transmission**

1. The calculated checksum will be merged with the original data/message.

2. The process of sending the message to the receiver starts.

Check Sum

- **Receiver's Side**

1. Break the bits of the message received from the sender into the 'k' number of blocks.

2. Now add all the blocks along with the checksum.

3. Perform the 1's complement to the result.

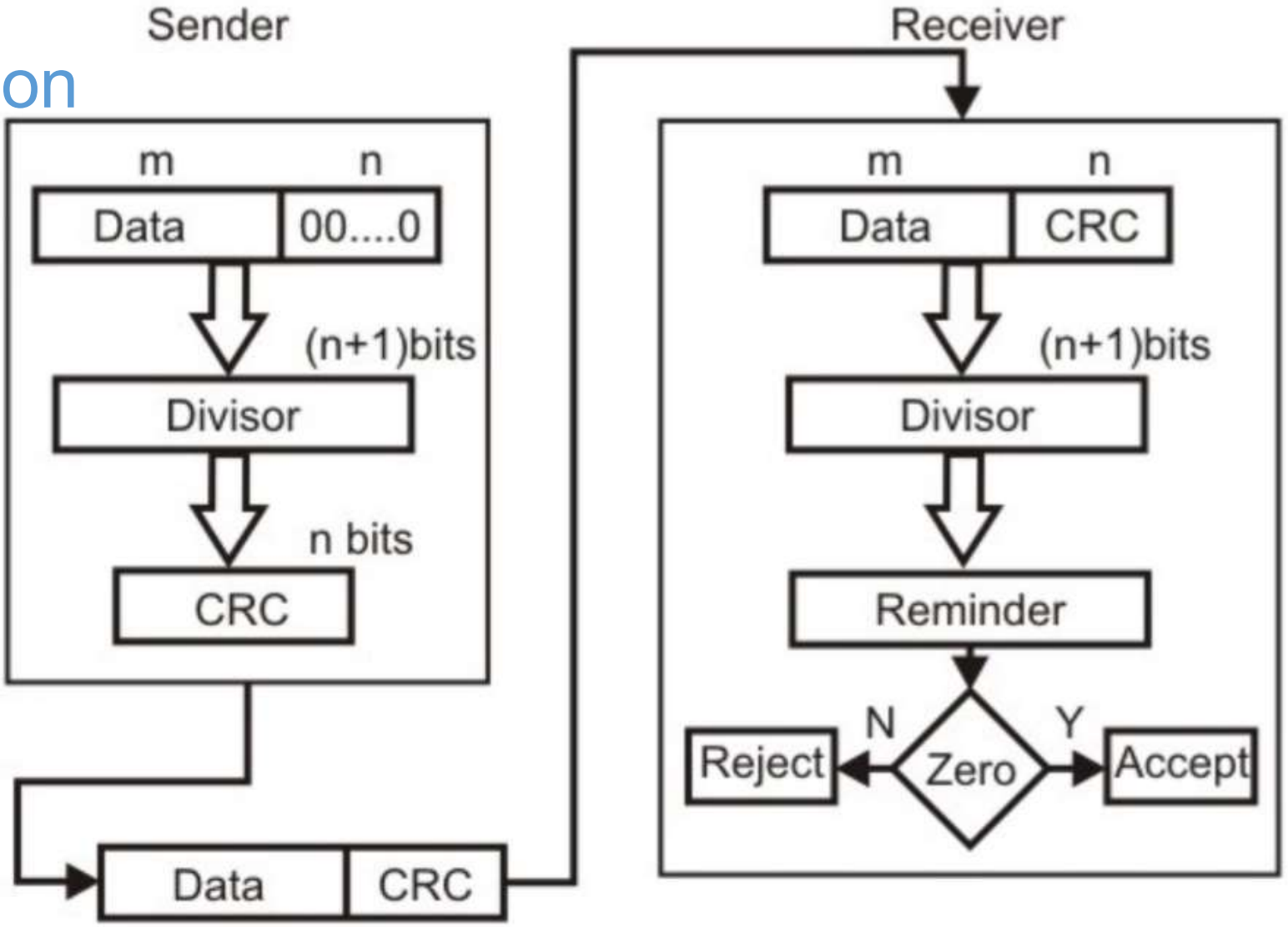
4. Now, there are two conditions:

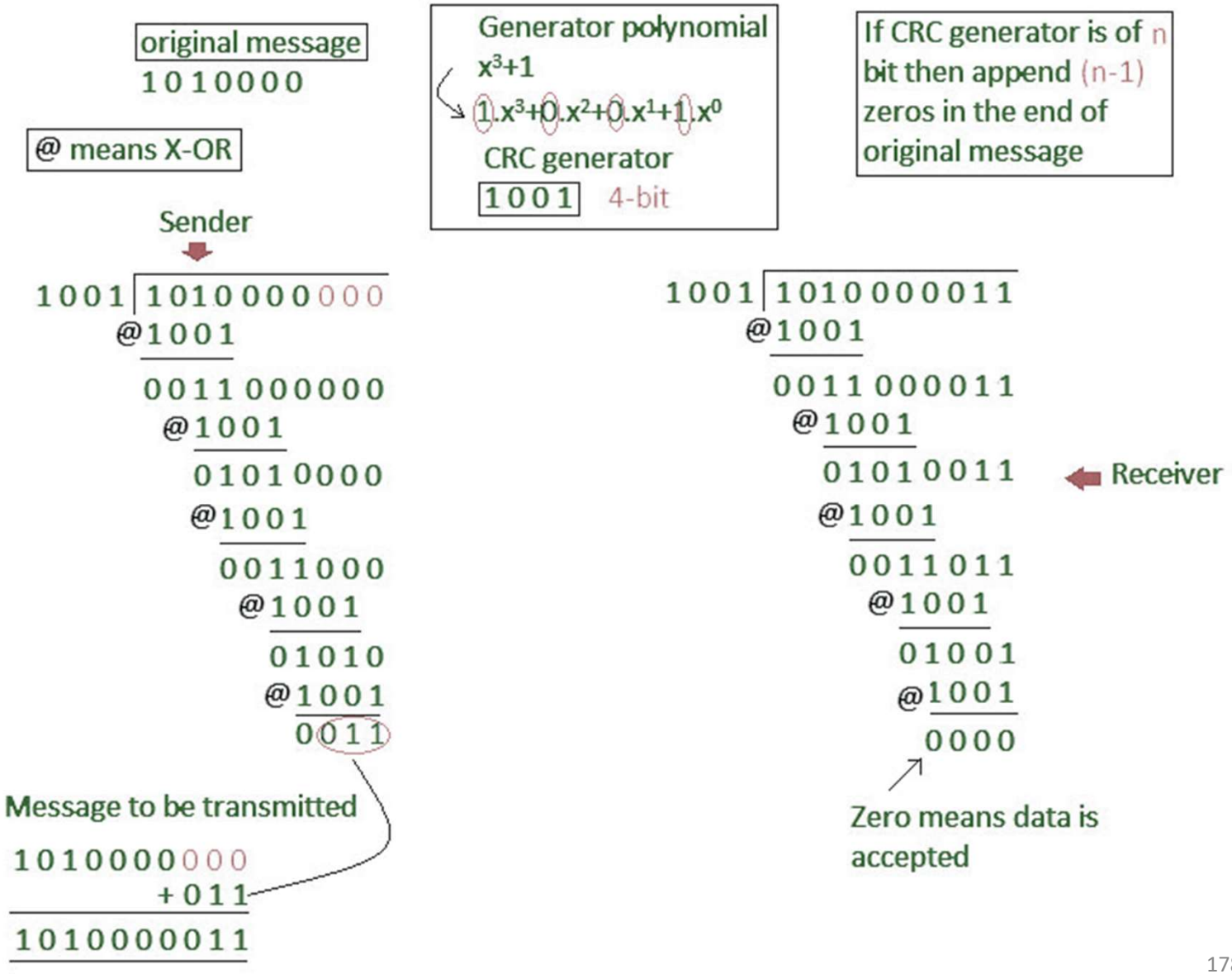
    1. If the result is 0, the message has no loss/ error.
    2. If the result is not 0, the message has a loss/error.

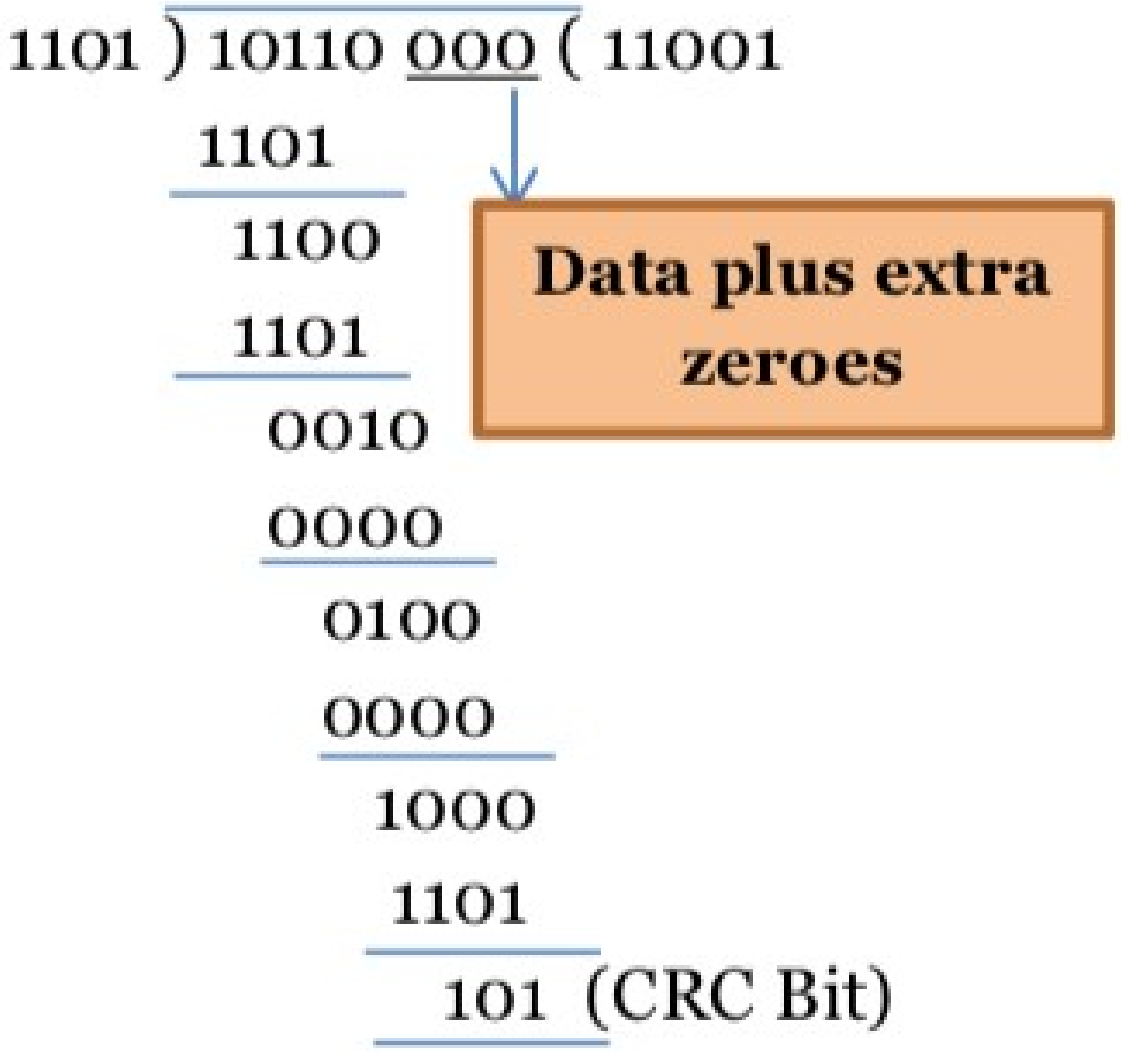
Cyclic Redundancy Code/Polynomial Code

- CRC was first proposed by W. Wesley Peterson, a mathematician and computer scientist, in 1961.
- CRC has advantages over other error detection methods in terms of
  - speed,
  - cost, and
  - correctness.

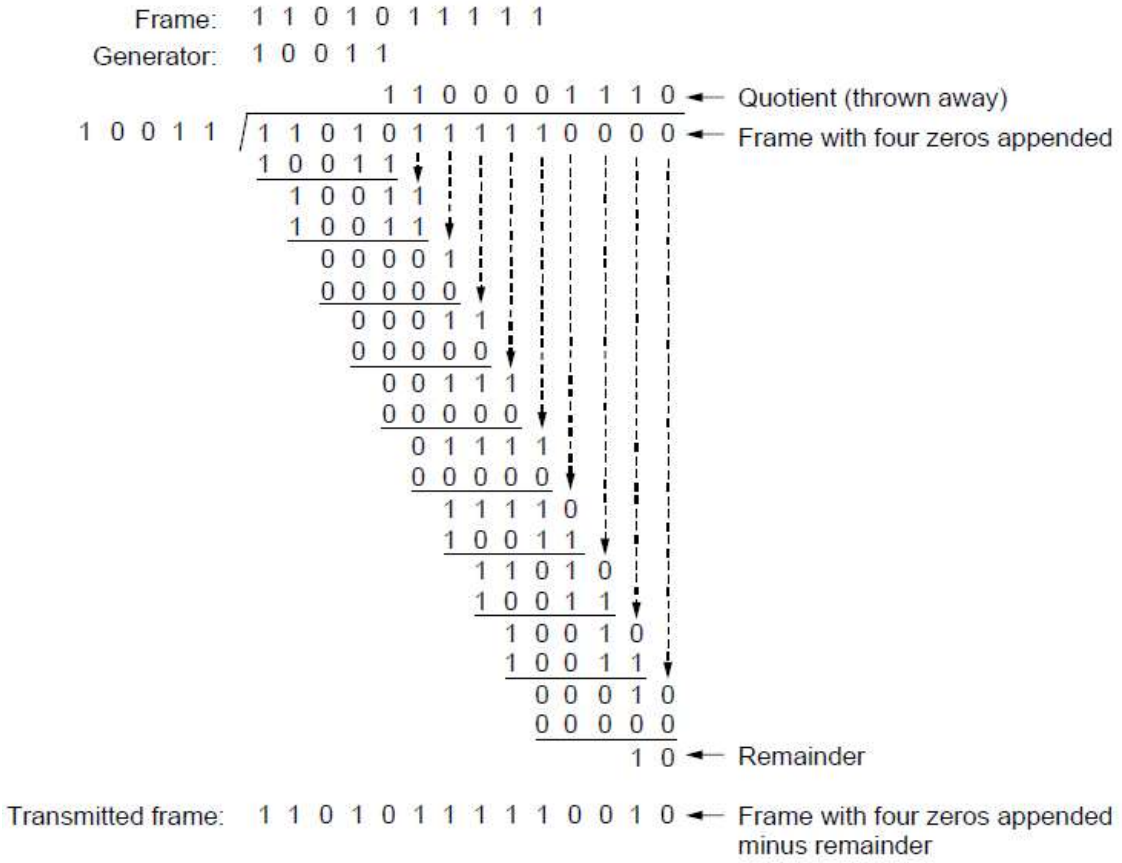
CRC Calculation




original message
1 0 1 0 0 0 0
@ means X-OR
Generator polynomial
x^3+1
$1.x^3+0.x^2+0.x^1+1.x^0$
CRC generator
1 0 0 1 4-bit
If CRC generator is of n bit then append (n-1) zeros in the end of original message
Sender
1 0 0 1 | 1 0 1 0 0 0 0 0 0 0
@1 0 0 1
0 0 1 1 0 0 0 0 0 0
@1 0 0 1
0 1 0 1 0 0 0 0
@1 0 0 1
0 0 1 1 0 0 0
@1 0 0 1
0 1 0 1 0
@1 0 0 1
0 0 1 1
Message to be transmitted
1 0 1 0 0 0 0 0 0 0
+ 0 1 1
1 0 1 0 0 0 0 0 1 1
1 0 0 1 | 1 0 1 0 0 0 0 0 1 1
@1 0 0 1
0 0 1 1 0 0 0 0 1 1
@1 0 0 1
0 1 0 1 0 0 1 1
@1 0 0 1
0 0 1 1 0 1 1
@1 0 0 1
0 1 0 0 1
@1 0 0 1
0 0 0 0
Receiver
Zero means data is accepted


1101) 10110 000 (11001
1101
1100
1101
0010
0000
0100
0000
1000
1101
101 (CRC Bit)
Data plus extra zeroes

Error-Detecting Codes (3)



Example calculation of the CRC

- Unit-1 completed